High-Stakes Performance Thresholds, Venue Choice, and the Market for Downhill Marathons*

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Abstract

Institutions that allocate scarce slots with a common performance threshold implicitly assume performance is comparable across the environments in which it is produced. When candidates choose those environments, the threshold misallocates slots—and creates a market for favorable venues. I study the Boston Marathon's qualifying system, where one public time standard applies at any certified marathon, using 17.2 million finish times and an AKM decomposition of times into runner and course effects, validated out of sample at Boston itself. Course effects span more than 20 minutes at the qualifying margin. The screening distortion is large and has doubled. Counting each linked Boston entrant once and using the effective admission cutoff, the share with no observed Boston-equivalent qualifying performance rises from 14.5% in 2004 to 28.5% in 2019 and 29.1% in 2025; the classifiable sample covers 40.0% of the 2025 field, and unmatched entrants are left unclassified. A venue-adjusted standard would displace about 13.0% of the field, far above what estimation noise can generate. Candidates respond: near-threshold runners sort to fast venues, the sorting grows after announced tightenings, and bunching tracks the moving standard on fast courses only. Venues respond: courses built to sell qualification entered and expanded until venue supply rivaled runner sorting as a source of misallocation. The same forces contaminate the value-added measure itself; I show how moving-threshold placebos, retests, and directional mover asymmetries detect and bound the contamination wherever fixed-effect performance measures are used for high-stakes selection.

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1 Introduction

Every fall, thousands of runners a few minutes too slow for the Boston Marathon travel to races built to fix that. Some descend four or five thousand feet through mountain canyons. Others are flat, small, and paced to the second. One series is named for the prize itself: at the "Last Chance BQ.2" marathons, the product—a Boston-qualifying time, a "BQ" in the sport's argot—is right there in the name. The strategy works. The share of Boston's field qualifying on courses that drop more steeply than Boston itself has nearly doubled since 2004, from 9 to 17%: runners, quite literally, racing to the bottom.

Boston is a clean case of a general problem. Scarce prizes—admission slots, scholarships, grades, promotions, ratings—are allocated by thresholds applied to measured performance, and the measurement is produced in environments candidates can choose: the courses behind a GPA, the sitting and form of a standardized test, the agency a security issuer pays for a rating. A common threshold looks neutral only when performances are comparable across those environments. When they are not, the threshold changes more than who clears a line: candidates move toward favorable venues, and suppliers build more of them. A nominally common standard creates a market for favorable measurement.

Boston makes this mechanism unusually visible. The race admits runners who beat an age–gender time standard at any certified marathon, so one public threshold applies at more than a thousand heterogeneous venues—some climb, some descend, some are exposed to heat or wind—while Boston's field is rationed by an effective cutoff that can be tighter than the posted standard. The threshold moves at announced dates and is valuable enough to shape years of training and travel. In 2025 the Boston Athletic Association acknowledged the comparability problem by announcing time penalties for courses with at least 1,500 feet of net elevation drop. The policy is a useful first step, but it also reveals the design question at the heart of the paper: how should a principal compare performances when both candidates and venues respond to the comparison rule?

I assemble 17.2 million marathon performances, link repeat runners across more than

1,000 courses, and use those movers to estimate an AKM-style decomposition—the two-way fixed-effects tool from labor economics, with runners as workers and courses as firms—of finish times into persistent runner ability, course speed, common year effects, age, and unusual race-day weather (Abowd, Kramarz and Margolis, 1999; Card, Heining and Kline, 2013). The intuition is simple. If the same runner is systematically faster at one course than another, after accounting for aging and weather shocks, the repeat-runner difference identifies a course advantage. A connected network of 2,631,055 runner identities places 1,162 courses on a common Boston-equivalent scale. Estimated course differences span more than twenty minutes at a qualifying pace.

The methodology is also part of the economic contribution. Permanent sorting of faster runners into faster courses is not, by itself, the main threat: runner fixed effects absorb persistent ability. The threat is transitory effort that depends on the chosen venue. A runner may peak at a qualifying race and later run Boston for the experience rather than for another personal best. A naive repeat-runner comparison then attributes part of the qualifying campaign to the course. The key diagnostic is direction: physical course differences are symmetric—the gap between courses A and B should not depend on which one came first—while peaking at a chosen qualifier is not. Section 3 develops the tests and the correction. The resulting behavior-adjusted indices differ most at the dedicated qualifying products and very little for the ordinary road courses used by most runners. The substantive results survive the full corrected-index band.

The behavioral evidence shows that the Boston incentive, rather than generic time goals, is driving venue choice. Runners close to their own qualifying threshold are increasingly likely to choose a fast course; runners far from the threshold are not. Finish-time densities bunch at the active BQ standard, and the spike moves when Boston moves the standard. Most importantly, the bunching is pronounced at fast and downhill courses but small at slow courses. The response therefore combines extra effort with deliberate venue selection. On the other side of the market, fast courses and explicitly branded qualifying products enter and expand to meet that demand.

The consequence for Boston's screen is large and rising. Among linked Boston entrants with an observed performance that clears the effective admission cutoff, I ask whether any observed qualifying-window performance would still clear after conversion to a Boston-equivalent time. The share for whom the answer is no rises from 14.5% in 2004 to 28.5% in 2019 and 29.1% in 2025. The classifiable sample covers 49.5%, 44.6%, and 40.0% of the complete Boston fields in those years; unmatched entrants remain unknown rather than being coded as unaffected. I retain 2026 as a provisional endpoint, but treat 2025 as the latest trusted cohort. A fixed-field counterfactual shows that the distortion changes not only labels but membership: a venue-adjusted rule replaces course-advantaged qualifiers with runners whose Boston-equivalent performances are faster.

The application illustrates a broader problem in measurement economics. Fixed effects are often used to recover the contribution of firms, schools, hospitals, judges, or neighborhoods from movers. When movement reflects persistent type, person effects can absorb the sorting. When agents choose an environment for a temporary campaign, treatment, test, or evaluation, the environment effect can absorb the campaign itself. Directional mover tests and routine-use comparison groups offer practical diagnostics. The two-sided-market response is equally general: schools, ratings agencies, and other evaluators can redesign the product that a threshold rewards (Sabot and Wakeman-Linn, 1991; Bolton, Freixas and Shapiro, 2012; Dranove et al., 2003).

The policy section compares four approaches: the current common cutoff, the BAA's one-sided net-drop penalties, a transparent symmetric physical adjustment, and an Empirical-Bayes course-effect handicap. Estimated course effects use the most information when a course has a sufficient history. A physical formula can cover new or sparsely observed courses, but it should both penalize favorable courses and credit difficult ones. The relevant comparison is among U.S. road courses and must be evaluated out of sample; trail races are informative about measurement but are a different policy universe.

Section 2 describes the rule, data, and selection mechanism. Section 3 explains the AKM estimator, its identifying assumptions, the targeted-effort bias, and the corrections. Section 4 presents runner demand, bunching, selection, and venue supply. Section 5 measures the rising distortion. Section 6 compares rules Boston could implement, and Section 7 draws the broader lessons.

2 The Boston Rule, Data, and Selection Mechanism

2.1 The Boston Marathon Qualifying System

The Boston Marathon is the world's oldest annual marathon and one of its most prestigious. Unlike most large-city marathons, Boston relies on time-based qualification rather than a lottery: runners must achieve an age- and gender-specific qualifying time at a certified marathon within a designated window. The threshold is high-stakes, clearly stated, and publicly known; as BAA President Jack Fleming puts it, runners "earn a spot on the starting line" (Boston Athletic Association, 2025*b*).

Regular increases in applications have made Boston substantially more selective. For men aged 18–34 the posted standard tightened from 3:10 (through Boston 2012) to 3:05 (2013–2019), then 3:00 (2020–2025), and now 2:55 (2026–27). Two dates matter for each tightening and the paper uses both. The *announcement* is the information event: on February 16, 2011, the BAA announced both the rolling fastest-first registration process (effective for the 2012 race) and the five-minute tightening of all standards (effective for 2013); the September 2018 and September 2024 announcements followed the same pattern. The *regime cutover* is the first race date whose qualifying window falls under the new standard (mid-September 2011, 2018, and 2024); exhibits that classify qualifying performances by the standard they faced use the cutover, while behavioral event studies use the announcement, since that is when runners could begin to reoptimize.[1]

[1] In oversubscribed years the *effective* cutoff is the posted standard minus an admission buffer, unknown

The qualifying system applies the same time cutoff across all certified marathons regardless of course difficulty: a 3:00 at the steeply downhill REVEL Big Bear and a 3:00 at the hilly Marine Corps Marathon are treated identically, though Section 3 estimates they reflect underlying performance differences of more than 20 minutes. In 2025 the BAA announced, for the 2027 race, net-elevation-drop penalties (5 minutes for 1,500–2,999 feet of net drop, 10 minutes for 3,000–5,999, and disqualification at 6,000+) (Boston Athletic Association, 2025*a*)—the first departure from the common-cutoff rule. Section 6 evaluates how much of the venue advantage it actually reaches.

2.2 Data

The data are a web-scraped panel of marathon results from race websites and aggregators (MarathonGuide.com, 2024): 17,206,046 finish-time observations at over 1,000 marathon courses in the US and 18 other countries, 2000–2026. A multi-stage entity-resolution procedure links runners across races by name, gender, and year of birth, erring toward conservative matching because false merges create spurious mobility links between courses while false splits only discard information. AKM estimation uses runners observed at least twice within their geographic region (11,142,687 observations on 2,631,055 regional runner identities at 1,162 courses); the 2.0 million runners observed on more than one course provide the cross-course comparisons that identify the course effects. Restricting to the highest-confidence identity matches leaves the course ranking essentially unchanged (Spearman rank correlation 0.95; Table D1). The selection and choice analyses additionally use a geocoded US-resident subpopulation of about 1.9 million runners. Appendix C details the entity resolution, sample construction, and time-variable selection, and Table C1 summarizes the samples and where each is used.

until registration closes, because Boston admits qualified applicants in order of their margin until the field fills. The posted standard is the target runners pursue and governs the behavioral analyses; the effective cutoff governs actual admission and enters the policy counterfactuals.

2.3 Definitions

The *posted* BQ threshold is the announced standard by age and gender; the *effective* cutoff subtracts the year-specific admission buffer. An *apparent qualifier* meets the posted standard; an *actual entrant* runs Boston in the following one or two years. The *neutral BQ margin* is the standard minus the runner's finish time after stripping the AKM course effect, so positive values mean the runner would qualify on a Boston-equivalent course. A *fast* course is at least 2% faster than Boston on the AKM log scale (about four minutes at a 3:20 pace); *very fast* is at least 4%. "Near-BQ" means close to the runner's own standard; exact bandwidths vary by exercise and are collected, with each exercise's rationale, in Appendix C (Table C2).[2]

The setting is a concrete instance of a general screening problem, and the paper uses both vocabularies interchangeably:

The screening vocabulary and its marathon counterparts

General object	Marathon counterpart
Agent / candidate	Runner
Evaluation environment / venue	Marathon course
Principal	Boston Athletic Association
Common threshold	BQ standard (age–gender qualifying time)
Venue supplier	Race organizer
Scarce slot	Boston entry
Venue value-added	Course speed effect

[2]Two global conventions hold throughout. Eras are defined by the standard in force for the relevant Boston cohort (pre-2013 / 2013–2019 / 2020+ for cohort exhibits) or by the announcement cutovers (pre/post September 2011 and 2018) for exhibits that classify qualifying performances by regime; each exhibit states which. COVID-era observations are handled uniformly: the cancelled Boston 2020 cohort is shaded or omitted, and the atypical Boston 2021 field is suppressed from take-up calculations.

2.4 How a Common Threshold Changes a Market

Observed performance combines four things a threshold cannot separate: the candidate's underlying ability, the systematic advantage of the environment they are evaluated in, the effort they bring, and luck. A principal who ranks candidates on that single number, against one common bar, inherits three consequences the paper documents in turn (cf. Baker, 1992). A common threshold confounds person and place, so equally capable candidates qualify at different rates purely by where they are measured. Tightening the threshold amplifies the venue advantage, because as the bar moves into the thin upper tail of the ability distribution a fixed environmental edge pushes more candidates across it. And the strategic response concentrates among marginal candidates, who have the most to gain from a favorable venue.

The framework identifies two linked distortions. The *screening distortion* arises whenever a common threshold is applied to heterogeneous venues, even absent strategic behavior. The *market distortion* is the endogenous response: candidates redirect venue choices, and venues themselves enter and grow, to exploit the advantage. The first does not require the second, but the second is induced by the first.

The supply side has an entry condition. A prospective venue pays a fixed cost to enter, offers a systematic advantage, and earns margins on the candidates it attracts. Its customers are the candidates for whom the advantage has option value: those within reach of the standard only at a favorable venue. Each tightening strands a dense band of aspirants just outside the new standard, raising the mass of candidates for whom a favorable venue restores qualification—so the return to entry rises exactly when the standard tightens, and venues built to supply the advantage should enter after announced tightenings. And because the rule rewards the advantage however produced, entrants will produce it in whatever form the rule does not police. Section 4.4 tests the timing prediction; Section 6 returns to the form-shifting one.

The same framework disciplines measurement. What an AKM regression recovers for each course is the *observed* venue value-added,

$$\psi_c^{\text{naive}} = \underbrace{\delta_c}_{\text{physical venue speed}} + \underbrace{S_c}_{\text{selection into venue}} + \underbrace{E_c}_{\text{target effort / peaking}} + \underbrace{O_c}_{\text{order / edge effects}} + \text{noise}, \tag{1}$$

because the runner effect absorbs each runner's *average* state but not the systematic within-runner effort, peaking, and ordering that correlate with venue choice. The component magnitudes are measured, not assumed (Section 3). One normative sentence guides the policy analysis: a venue-adjusted standard should neutralize δ_c—gravity is not merit—but not effort, since training to peak at a goal race is what the institution means to reward.

3 Measuring Course Speed: AKM, Bias, and Correction

This section builds the measurement tool: a course-speed index estimated on the full panel. Section 3.3 then asks, assumption by assumption, whether the tool should be believed—and finds that the one assumption that fails does so in a way that is itself the subject of the paper.

3.1 Using Repeat Runners to Compare Courses

I estimate

$$\ln T_{irt} = \alpha_i + \delta_r + \gamma_t + f(\text{age}_{it}, \text{male}_i) + g(W_{rt}) + \varepsilon_{irt}, \tag{2}$$

where T_{irt} is the finish time of runner i in race r in year t, α_i is a runner effect, δ_r a course effect, γ_t a year effect, f an age–gender polynomial, and g a quadratic in demeaned race-day temperature. Identities are defined within five geographic regions to limit false links; estimates are normalized to Boston ($\hat{\delta}_{\text{Boston}} = 0$) using about 24,000 bridge runners, shrunk by Empirical Bayes—noisy effects are pulled toward the mean so sparsely observed courses

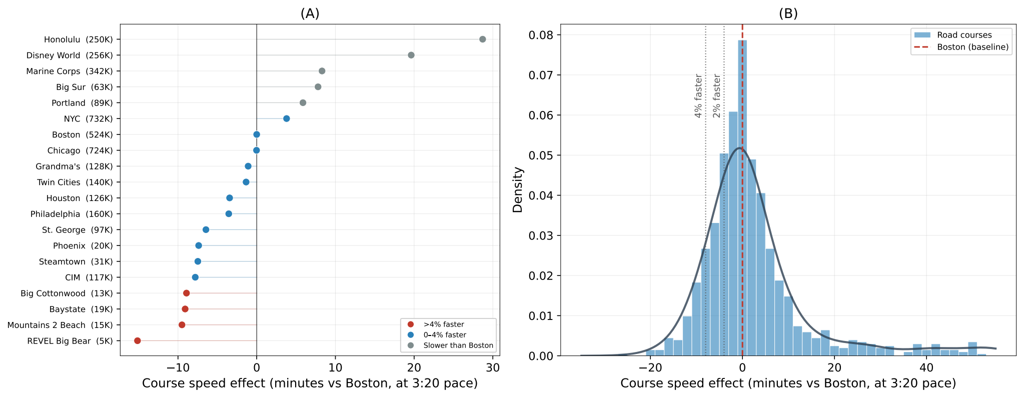


Figure 1: How much marathon courses differ. Panel A reports recognizable courses in minutes relative to Boston at a 3:20 pace. Panel B shows the distribution across road marathons; negative values are faster than Boston.

are not over-interpreted—and published for the 1057 courses with at least 50 observations. Throughout, $\hat{\delta}$ denotes this published, shrunk index unless a variant is named. Appendix B gives the full specification, the progressive R^2 decomposition, the shrinkage estimator, and an algebraic invariance result (the fixed-field displacement of Section 5 is invariant to any common shift of the index, so only cross-course *differential* error can matter for the headline); Appendix D reports sample-restriction robustness.

3.2 How Much Courses Differ

Figure 1 shows the result. Purpose-built downhill courses run 4–9% faster than Boston; major city races and trail marathons run slower; estimated effects span more than 20 minutes for runners at the qualifying margin (the full published index is Appendix H). Most courses are *slower* than Boston, so most BQ pursuit happens at a venue disadvantage—a fact Section 5 returns to when it prices the reallocation.

Two facts anchor the index to physical reality. First, physical predictors work in the expected direction, but most of their apparent power comes from trail and mountain races, where gross climbing alone has a cross-validated R^2 of 0.85. In the policy-relevant sample of 439 profiled U.S. road courses, the course-weighted out-of-sample R^2 rises from 0.07 for net drop to 0.33 for gross climbing and 0.47 for the full physical-and-weather formula—and even that formula explains only 0.12 of the variation weighted by where runners actually race, a ceiling that correcting for the AKM target's own sampling noise (repeat-sample reliability 0.96) raises only to 0.49. A symmetric physical formula is therefore a useful fallback for new courses, not a substitute for repeat-runner course effects. Nor is the drop in predictor fit a failure of the AKM index: re-estimating the model after excluding trails leaves the remaining course effects almost unchanged (Spearman $\rho = 0.999$; mean absolute change 0.2 minutes; profile coverage and detail in Appendix B). Second, the median within-course year-to-year standard deviation of course means is 2.8 minutes, small against a cross-course range of more than 40 minutes (Appendix E). The sharpest validation—the same runners retested across courses—belongs to the next section, because it is also the test of the model's central assumption.

3.3 When Repeat-Runner Comparisons Are Credible

Equation (2) is the workhorse of modern measurement economics: the same additive person-plus-environment decomposition as AKM wage models (Abowd, Kramarz and Margolis, 1999; Card, Heining and Kline, 2013) and teacher value-added (Chetty, Friedman and Rockoff, 2014), with runners as workers and courses as firms (Bonhomme, Lamadon and Manresa, 2026, provide a user's guide to the wage-model version). Everything the device reports rests on three assumptions. In most applications they must be defended indirectly, because the environments are hard to characterize and the movers are few; here each can be put to the data directly. This section asks the three questions in turn. Two receive unusually clean yeses. The third receives a no—and the anatomy of that no organizes the rest of the paper.

3.3.1 Are course effects stable and additive?

Additive separability says a course shifts every runner's log time by the same amount. It is the assumption every fixed-effects application waves through, because testing it requires observing the same person in many environments; here, millions of repeat-runner pairs test it directly. For each same-runner pair of races, the change in log finish time should equal the difference in the two courses' estimated effects, one for one. It does: regressing within-runner time changes on course-effect differences gives $\hat{\beta} = 0.933$ across all 10,934,525 cross-race pairs, and the Boston retest—pairing each Boston run with the same runner's prior marathons within 18 months, so that the index estimated elsewhere is tested out of sample at the prize venue—gives $\hat{\beta} = 0.851$ on 690,058 pairs (Figure 2; Appendix B, Table B1). Nor does the validation rest on Boston-destination pairs, the most campaign-selected population in the data: rerunning the retest with lottery-entry majors as the destination gives slopes of 0.99 (Chicago), 1.00 (New York), and 0.83 (Marine Corps) (Appendix Table B4), so the transfer relationship is not an artifact of the Boston-bound population alone.

The slopes are close to one, and their small deviations are informative. False entity matches among common names compare different people and push the slope above one; restricting to medium-frequency names with low-confidence links removed brings it to 0.98 overall and 0.93 into Boston (Appendix Table B2). Conversely, slopes into Boston are lower than slopes elsewhere: a performance peaked at the qualifying race does not transfer fully to the goal race. Additivity is therefore a good approximation, while its signed residuals identify the two biases the paper treats below.

3.3.2 Does the mover network connect the courses?

The second assumption is enough movement to separate person from place. Here 1,985,267 runners are observed on more than one course, connecting the 1,162 courses—about 1,708 movers per course—so the mover graph is unusually dense. I nevertheless avoid variance-decomposition claims, shrink noisy course effects toward Boston with Empirical Bayes, and carry a conservative index that sets statistically imprecise effects to zero. Leave-out estimates, noise placebos, and runner-clustered bootstrap intervals check that remaining sampling error cannot explain the policy results. Finally, the estimand is a course *as historically run*: route changes and young courses motivate regular re-estimation rather than interpreting

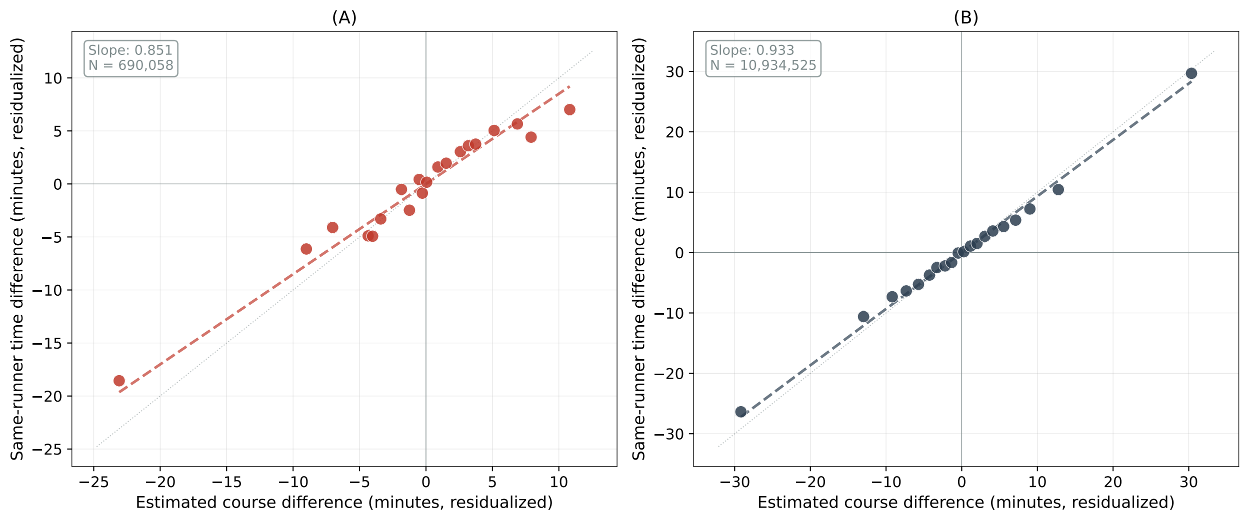


Figure 2: Why repeat runners identify course speed. For the same runner, the time difference between two races should equal the estimated difference between their courses. Panel A uses Boston as a common retest; Panel B uses all cross-race pairs. Points are pair-weighted quantile-bin means with age and gender partialled out of both axes; the dashed line is the pair-weighted regression slope of Table B1; the dotted line is the one-for-one benchmark. The Boston slope sits below the all-pairs slope—the signature of peaking at the qualifier that Section 3.4 measures.

each effect as a physical constant.

3.3.3 Does targeted effort contaminate the comparison? Yes.

The third assumption is that movement is unrelated to the *transitory* part of performance. Persistent sorting of strong runners into fast courses is not the problem: runner fixed effects absorb persistent ability. The problem is differential temporary state. Runners choose qualifying venues for the attempts when they are trained, tapered, healthy, and close to their standard. A naive course effect can then combine physical venue speed with a qualifying campaign.

This is the Rothstein critique of value-added measurement in unusually visible form (Rothstein, 2010; Chetty, Friedman and Rockoff, 2014). The incentive is a public threshold that moves on announced dates; venues are discrete and physically measurable; weather is observed; and the same runner appears on both sides of a switch. Contamination should therefore be directional, concentrated near the threshold, and strongest at qualifying venues. The data show all three: runners under-transfer from qualifiers into Boston, while reverse moves and lottery-entry destination races do not show the same pattern. The failure of exogenous mobility is thus measurable behavior, not an unstructured nuisance. The repeat-runner design remains a valid measurement device once the venue-specific share of strategic attempts is used to scale the correction.

The organizing result is proportionality. Behavioral contamination can enter a course effect only through runners who use that course strategically. Let s_c denote the share of course c's finishers whose qualification depends on the course. To first order,

$$\psi_c^{\text{naive}} \;=\; \delta_c \;+\; s_c\, b_c \;+\; \text{noise}, \tag{3}$$

where δ_c is physical course speed and b_c is the average behavioral residual of strategic users. Equation (3) collapses the non-physical components of equation (1): persistent selection is already absorbed by the runner effect, and the remaining within-runner selection, effort, and order terms $(S_c + E_c + O_c)$ operate only through strategic users, which gives $s_c b_c$ to first order. The median course has a strategic share of about 1.0%. Contamination is therefore concentrated at the dedicated qualifying products rather than spread uniformly across the course index. Directional pairs estimate b_c; observed strategic shares supply s_c; routine attempts and reverse pairs provide independent checks.

3.4 Order Effects, Edges, and Peaking

For an ordered same-runner pair $r \to s$, define the residual $\eta_{r\to s} = (\ln T_s - \ln T_r) - (\hat{\psi}_s - \hat{\psi}_r)$. Its mean is near zero, but it is systematically positive *into* Boston: runners are +7.9 minutes slower there than the additive model predicts, versus -2.1 minutes when Boston is the origin—an asymmetry of +10.1 minutes (Appendix Figure B1a and Table B5). Physical course differences should reverse sign when the order reverses; peaking at the qualifier does not.

The asymmetry is absent for lottery-entry destination races and is largest when the runner qualified at the origin, exactly as targeted effort predicts. Directional mover residuals provide the course-level estimate of that behavioral component.

3.5 Strategic Shares and the Behavior-Adjusted Index

The strategic share is the proportionality constant in equation (3), and measuring it shows where contamination can and cannot live. At St. George, a steeply downhill course, course-dependent qualifiers are about 5.6% of finishers; the median among courses with a measured peaking residual is 1.0%; only the dedicated qualifying products reach contamination-relevant shares (the fastest courses' shares are tabulated in Appendix Table B6). Because strategic users are a minority almost everywhere, they barely move the course effect: re-estimating the model after excluding near-threshold observations (the "clean" index) leaves the benchmark downhill courses essentially unchanged. A steeply downhill course is genuinely fast for everyone who runs it.

The behavior-adjusted index implements equation (3), netting out the course-level peaking contamination by scaling the origin peaking residual η_c by the strategic share:

$$\hat{\delta}_c^{\text{adj}} \;=\; \hat{\delta}_c \;+\; s_c\,\eta_c. \tag{4}$$

The into-Boston residual η_c estimates the negative of b_c in equation (3)—a course whose users peaked there looks too fast, and η_c is positive—so adding $s_c\,\eta_c$ undoes the contamination term $s_c\,b_c$. The adjustment is deliberately modest and agrees with the clean re-estimate—St. George moves from -6.4 to -5.7 minutes relative to Boston, and the typical course barely moves. Only a course's strategic users additionally peak, so only their share of the field is netted out. Applying the raw retest residual to the whole course instead would be a selection error: the residual is estimated from the self-selected minority who qualified and went to Boston, and attributing their behavior to every finisher would flip genuinely fast courses to slower-than-Boston. Section 5 uses the behavior-adjusted index as the headline screen, with the full suite as the robustness band.

The correction is concentrated where the model says it should be: it is monotone in the strategic share, exceeds one minute only at high-strategic- share courses, and is below one minute at courses accounting for 99% of U.S. road finishes (Appendix Figure A7). Thus the naive and physical indices are nearly identical for ordinary venues; the meaningful exceptions are the qualifying products themselves.

3.6 Composition Controls: Routine Attempts and Reverse Pairs

Two independent composition checks reach the same conclusion. A *routine-attempt* index uses local or repeatedly visited races, removing one-time goal-race visits; the visitor premium concentrates at dedicated qualifying venues and is near zero at ordinary majors. A *reverse-pair* index averages both directions of the same course pair, differencing out order-specific peaking. Construction and course examples are in Appendix B.6.

3.7 The Corrected-Index Suite

The policy conclusion does not hinge on a unique "true" index. Across all 7 indices, built on one consistent sample, displacement ranges from 9.7% to 14.0%; the policy accounting of Section 5 reports its headline against this band. Table 1 collects the main indices, what each neutralizes, and the fixed-field displacement each implies. Three further variants—the clean re-estimate excluding the threshold region, the reverse-pair symmetric estimator, and a conservative shrinkage that zeroes statistically imprecise courses—appear in Appendix Table D7. Two members of the suite are directly implementable as rules—the conservative shrinkage and the observable-based formula, which needs no race history and so covers brand-new courses; Appendix B.6 details both and the practical hybrid that combines them. One limitation qualifies the band's interpretation: the observable-predicted index is fit to contaminated fixed effects, and peaking correlates with fast terrain, so it is not a fully behavior-free benchmark either—the band, not any single index, is the answer.

Table 1: The course-speed index suite: construction, what each neutralizes, and the implied fixed-field displacement (US 2018–19 qualifiers, field size 84,925). Appendix Table D7 reports the full 7-index suite.

Index	Neutralizes	Displaced (%)
Behavior-adjusted (headline), eq. (4)	Course-level peaking of strategic users	13.0
Naive AKM ψ (starting point)	—	13.7
Routine-attempt	Visitor/goal-race composition	11.2
Observable-based (terrain)	All non-physical variation (and some physical)	9.7

Notes: The four main indices; the remaining three (clean re-estimate, reverse-pair symmetric, conservative shrinkage) appear with these in Appendix Table D7. Displacement is the one-sided share of the fixed field reallocated when candidates are ranked by the index-adjusted margin.

4 How Runners and Venues Respond

Section 3.3 ended with an assumption that fails; this section documents the behavior that breaks it—who sorts, when, how much, and through which margin—by contrasting populations that face the qualifying incentive with populations that do not: near-threshold versus inframarginal runners, BQ thresholds versus round-number placebos, announced tightenings versus surrounding years. Every estimate uses only the coarse fast/slow course classification, which is robust to all of the index corrections in Section 3; because the estimates measure deliberately different estimands on different populations, Appendix Table D2 maps each to its object.

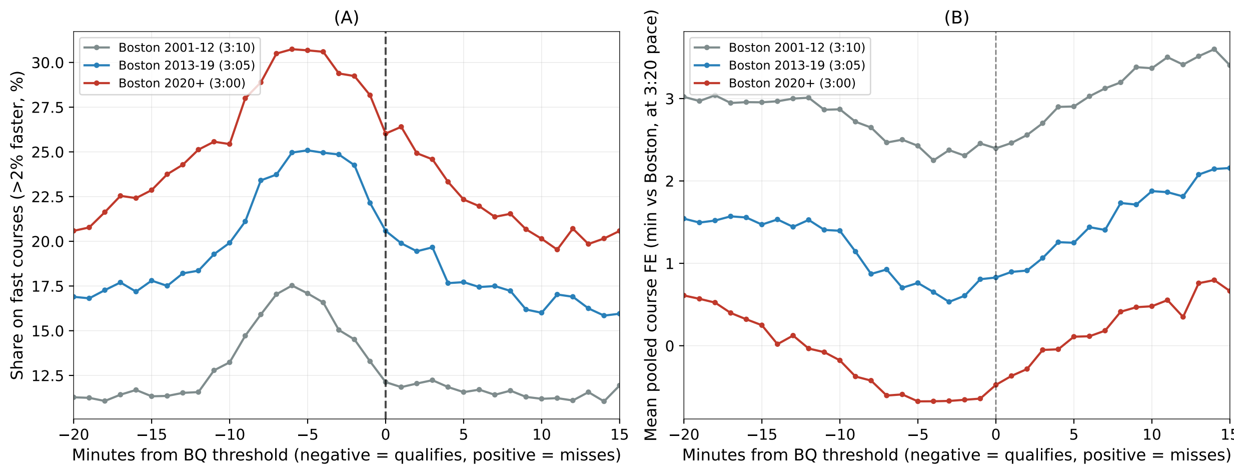


Figure 3: Runners near the Boston threshold increasingly choose fast courses. Panel A reports the fraction choosing a fast course by Boston-equivalent minutes from the runner's standard. Panel B reports the mean course advantage. The near-threshold gradient grows across eras.

4.1 Sorting toward faster courses near the margin

Runners near the qualifying threshold concentrate on faster courses, and the concentration has grown. Figure 3 plots the fast-course share against the neutral BQ margin by era. The share is flat far from the threshold and rises sharply near it; the near-threshold excess grew era by era—4.6, 7.9, and 10.9 percentage points relative to each era's field in the 3:10, 3:05, and 3:00 eras. Inframarginal runners—too far from the cutoff for course choice to matter—show no comparable gradient, which addresses amenity- and geography-based alternatives: both groups face the same courses, travel costs, and calendars; only one faces the incentive.

Treating the inframarginal venue distribution as the no-incentive counterfactual, roughly one in five near-BQ runners is at a course they would not otherwise have chosen (Appendix Figure D5).

4.2 Predictable pressure and the announcement design

Selection is forecastable from pre-race information—the Chetty, Friedman and Rockoff (2014) forecast-bias logic run in reverse, with predictable selection as the outcome of interest. Runners under more pre-race BQ pressure choose measurably faster venues, and the fast-course share peaks for runners whose prior best just missed the standard (Appendix Table D3).

The announced tightenings give the design-based version. In an event study of near-BQ runners' differential fast-course choice by year, the differential is flat before the February 2011 announcement and rises steadily afterward (Appendix Figure A1); the compact difference-in-differences coefficient is +3.2 percentage points (SE 0.16) and survives home-state and year fixed effects. Because a fast course lies within 150 miles of 77% of runners, geographic access does not gate the response.

A conditional-logit model of venue selection (US runners, US courses; Appendix G) prices the same response in choice terms: utility falls in a fixed long-distance travel cost and rises in the interaction of pre-race BQ pressure with course speed. The policy-relevant translation is not the average shift in chosen-course speed—mechanically small—but the change in qualification probability. Switching off the pressure channel and re-solving removes about 11.7 qualifications per 1,000 near-BQ runners, concentrated within a few minutes of the threshold, holding the course menu fixed. The model's role is corroborative; the design-based estimates above carry the identification weight.

Qualifying standards also relax discontinuously at five-year age boundaries—a notch that in principle offers two further response margins, venue choice and attempt timing. Neither is informative here: year-of-birth data cannot classify the boundary-adjacent cells where a venue response would appear, and the attempt-retiming test returns a precise null (Appendix F). No exhibit rests on either margin.

4.3 Bunching: effort and venue choice jointly

Marathon finishers bunch at salient round numbers (Allen et al., 2017). Boston-qualifying times are not round for most demographics, providing a sharp test of whether the threshold itself is a target. Three facts establish that it is, and that the response runs through fast venues. First, the bunching spike physically tracks the standard as the BAA moves it: recentering every demographic's finishes on its own pre-2011 standard, the fast-course density kinks exactly at the active BQ in every era, five minutes further left after each tightening, while the slow-course density stays smooth (Figure 4). A behavioral spike that follows an administratively moved cutoff cannot be a fixed feature of the environment. Second, the headline gradient: regressing each marathon's estimated BQ-bunching premium on its course speed, the slope is -0.0131 per minute of course speed after the 2011 announcement (permutation $p < 0.001$, reassigning course speeds across marathons) versus -0.0088 before (Appendix Figure A2)—the course-speed gradient in BQ targeting is a post-tightening phenomenon. Third, excess-mass estimates—the standard bunching estimator, which measures the extra density piled against the threshold relative to a smooth counterfactual (Saez, 2010; Kleven and Waseem, 2013)—by course type and era (Appendix Table D4) show the fast$-$slow excess-mass gradient at the BQ threshold is about three times its round-number-placebo counterpart (Appendix Figure F2); the cell-level interaction versions of these contrasts are signed identically but individually imprecise, and the marathon-level gradient above carries the inferential weight.

Finally, the sorting, bunching, and peaking facts move together when ordered by a single constructed pre-race BQ-pressure index, with one exception—the peaking-by-origin cross-section is dominated by course composition—detailed in Appendix F.

The demand response has direct analogs outside sport. Sorting toward favorable venues near a threshold is the running-market version of students selecting easier-grading departments (Sabot and Wakeman-Linn, 1991), and bunching that physically tracks an administratively moved cutoff is the same signature that accountability-threshold reforms and regraded examinations provide in education (Dee et al., 2019; Diamond and Persson, 2016).

4.4 Venue Supply Expands to Meet Qualifying Demand

Venue entry is the side of the feedback loop least visible in prior work; this section measures it. Fast courses expanded 2.3-fold, purpose-built qualifying products entered on a timeline

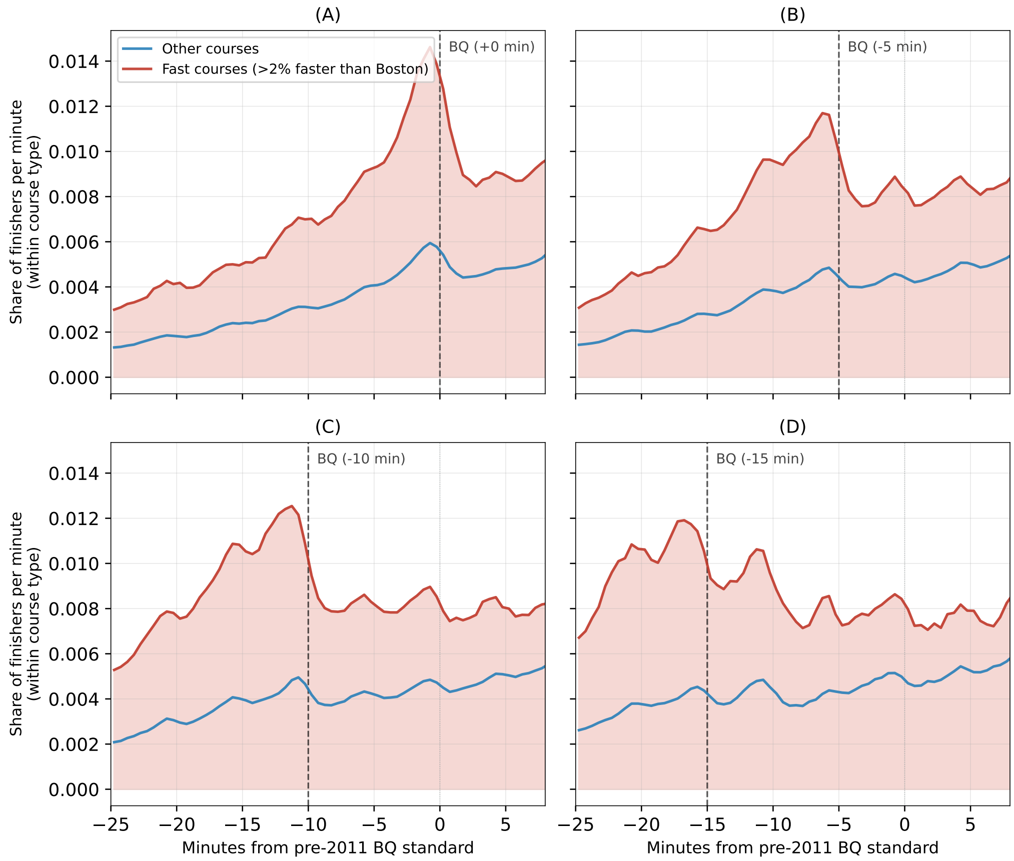


Figure 4: BQ bunching is concentrated at fast courses. Each panel shows one qualifying era, with finish times re-centered on the demographic's pre-2011 standard, so the successive standards appear at offsets of 0, −5, −10, and −15 minutes. In every era the fast-course density spikes at the standard then in force and the spike moves with each five-minute tightening, while the density at other courses remains comparatively smooth.

aligned with announced tightenings, and by the 2019 cohort the supply channel contributes 4.5 percentage points of the misallocation rate—the share of qualifiers whose qualification depends on their course, measured in Section 5—of the same order as candidate sorting.

4.4.1 Entry and growth of fast venues

The number of US courses at least five minutes faster than Boston at the M18–34 standard (about 3%—a stricter bar than the 2% "fast" classification used elsewhere in the paper) rose from 23 in 2004 to 54 in 2019, with the steepest growth between the 2011 and 2018 announcements (Appendix Figure A3, Panel A). The fast-course share of US finishers rose from

5.7% to 13.1% over the same period (Panel B). Entry concentrates in the product category the rule rewards: the REVEL series (purpose-built point-to-point downhill races, multiple launches 2014–2018) and the "BQ.2" races (Spring Chance, Last Chance Chicagoland, Last Chance Grand Rapids), whose branding—"BQ" is the product name—markets the screening rule itself.[3]

Entry is the cleanest supply margin, and Appendix Figure A4 shows its timing and composition. The fast share of newly entering US courses rises by roughly half, from about 18% of entrants before 2008 to about 26% afterward—peaking at 29% in 2011–2018—with the very-fast share more than doubling, from 5.5% to 14.1%. The flagship launches land almost entirely between the two announcements—nine purpose-built products enter from Big Cottonwood (2012) to Spring Chance BQ.2 (2018); Appendix Figure A4 names each. Entry selection toward fast products begins around 2008, after Boston's registration sellouts made the demand pressure visible and before the announcement that codified it—consistent with entry responding to the incentive, of which the announcement is the public signal.

The consequence for the screen is summarized by one statistic: in the Boston 2019 cohort, 41% of course-dependent qualifiers—those who clear the standard only by virtue of their course—ran a course that did not exist in 2010, versus 18% of all apparent qualifiers. These are equilibrium quantities, not exogenous supply shocks: entry responds to the same incentive that moves runners, and the decomposition of Section 5.3 accordingly treats supply and sorting as accounting channels with a common upstream driver.

4.4.2 Which courses over-attract marginal runners

A course-level diagnostic identifies who serves the marginal demand. For each course-cohort cell, I compute the share of finishers within ±3% of their standard and benchmark it against a smooth in course speed: the residual is the course's over- or under-attraction of marginal runners beyond what its speed predicts. The top over-attractors are the "last chance"

[3] The series in Appendix Figure A3 uses the five-minute definition; the 2% classification used elsewhere in the paper shows the same proportional growth.

BQ events—Last Chance BQ.2 Grand Rapids, Last Chance BQ.2 Chicagoland, and Spring Chance BQ.2—whose marginal-runner shares sit roughly double what course speed alone would predict (Appendix Figure D8). The venues created for marginal BQ chasers are the ones the diagnostic flags.

4.4.3 Take-up as revealed intent

Venues created to sell qualification convert qualifiers into Boston entrants at several times the take-up rate of the majors—revealed BQ intent on the demand side, and the product working as designed on the supply side (Section 5 reports the take-up accounting). By the Boston 2019 cohort the supply contribution to the misallocation rate (4.5 pp) is of the same order as the demand contribution (6.6 pp) under every counterfactual baseline (Section 5.3): the market distortion is two-sided.

4.4.4 The 2027 net-drop rule as an out-of-sample test

The BAA's announcement of net-drop penalties for the 2027 race (June 16, 2025; effective with the qualifying window opening that September) is the loop's next iteration, and it arrived at the very end of the sample period, so the data cannot yet show an entry response. What they can show is the target the next round will aim at. Of the 75 fast US courses active in 2024–25, 56—about 75%—fall below the 1,500-foot net-drop floor and are untouched by the rule, led by the fast-flat qualifying products: Spring Chance BQ.2 (essentially no net drop, yet a speed advantage around 16 minutes), BQ.2 Chicagoland, and the tunnel-family point-to-point races. The framework's form-shifting prediction is that entry, course design, and marketing pivot toward this unpenalized tier. Section 6 develops the rule's screening properties; the entry timeline around the 2011 and 2018 announcements (Appendix Figure A4) is the template against which the 2025 announcement will be read as the 2026–27 windows accumulate.

The supply response, too, has analogs wherever passage of a threshold is worth money or status: test-prep franchises, easy-grading course providers, diploma mills, and lenient raters are all entry on the same margin (Bolton, Freixas and Shapiro, 2012). Static audits of candidate gaming miss what may be the larger half of the equilibrium; in this setting venue supply grew to the same order as candidate sorting within a decade.

5 The Rising Distortion

By the most recent trusted cohort, nearly three in ten linked Boston entrants with an observed effective qualifying performance have no observed performance that would still qualify after Boston-equivalent adjustment. This section measures that distortion: its scale and growth; its decomposition into the formal rule, venue supply, and runner sorting; the separate participation-selection evidence; and the counterfactual field under a venue-adjusted standard. The benchmark is the principal's own stated objective—select the fastest runners on a common basis ("earn a spot")—so misallocation here means deviation from that meritocratic objective, not a social-welfare claim; the BAA may also value revenue or geographic breadth, which would change the normative reading but not the positive results. All accounting in this section runs on the US road-course universe.

The primary participant measure follows the administrative rule directly. For Boston entrant i in cohort Y, let c_{gY}^{eff} be the posted age–gender standard minus that cohort's admission buffer. The entrant enters the classifiable sample if at least one observed qualifying-window performance clears c_{gY}^{eff}. I classify the entrant as course-dependent when

$$\min_{j\in\mathcal{J}_{iY}}\{T_{ij}e^{-\hat{\delta}_{r(j)}}\} > c_{gY}^{eff},$$

so none of the entrant's observed performances would clear the effective cutoff on a Boston-equivalent course. This uses every observed performance in the window, not only the fastest or any selectively chosen race, and counts each entrant once. BQ age is age on Boston race day. Entrants with an unindexed observed performance are reported as unclassified. A separate performance-level series uses the posted standard to describe the larger population responding to the public target; it should not be read as the realized Boston field.

5.1 The distortion is large and has doubled

Figure 5 plots the posted-standard performance series alongside the direct entrant-level effective-cutoff series. The former rises from about 9.7% in the Boston 2004 cohort to 18.4% in 2019. Among linked, classifiable Boston entrants, the effective-cutoff rate rises from 14.5% in 2004 to 28.5% in 2019 and 29.1% in 2025, the most recent trusted endpoint—essentially all of the growth arrives by 2019, and the rate has since held at its elevated level. The classifiable sample covers 49.5%, 44.6%, and 40.0% of the observed Boston fields in 2004, 2019, and 2025; unmatched entrants remain unclassified rather than being coded as unaffected. Boston 2026 is retained at 29.7%, but is marked as provisional because its external course-level linkage diagnostic is less uniform; its classifiable sample covers 37.2% of the full field.

Neither index choice nor linkage composition drives the pattern. Re-classifying every entrant with the behavior-adjusted index moves the series only slightly, from 13.5% to 27.3%. The terrain-based observable index—which attributes to the venue only what physical features predict—puts the level lower (8.2% to 15.8%) but preserves the doubling. And within the linkage-homogeneous US-resident subpopulation, where match rates are uniform across cohorts, the series is if anything slightly steeper (14.6% to 30.8%), so the decline in classifiable coverage is not manufacturing the trend. The *level* depends on how much of a course's advantage the index attributes to the venue; the doubling does not. The shift is visible in the terrain itself: the share of the Boston field qualifying on any net-downhill course rose from about 30 to 40%, and on courses dropping more steeply than Boston itself from 9 to 17%.

5.2 Qualifying mechanics and the shrinking menu

Two mechanical facts explain why the distortion grows with every tightening. Each runner faces a menu of courses at which their ability can clear the standard, and the menu contracts sharply as the standard tightens: a runner with a neutral-course baseline of 195 minutes could qualify at 196 indexed courses under the 3:10 standard, 52 under 3:05, and 12 under 3:00 (Appendix Figure D4, Panel A). And tightening is not neutral across venues: on slow courses the cutoff sits in the thin right tail of the ability distribution and a five-minute tightening disqualifies few runners; on fast courses—where qualification rates run roughly 35–56% at the flagship purpose-built products, several times the 10–12% rates at the majors—it crosses dense regions and disqualifies many (Panel B), so the fast-course share of qualifiers rises mechanically with every tightening, before any behavior.

The growth could therefore come from the rule itself, from the changing venue menu, from candidate sorting, or from who claims a slot. The decomposition separates the channels.

5.3 Decomposing the growth: rule, supply, and sorting

To separate the pre-participation channels, I build a chain of counterfactuals that starts from random assignment of each Boston-cohort runner pool to a frozen 2005–07 course market at the 2005 effective cutoff, then progressively adds (step 2) the cohort's effective cutoff, (step 3) the cohort's actual course menu, and (step 4) the actual assignment of runners to courses. Step 5 is not an additional additive channel: it replaces performance-level accounting with the direct entrant-level measure above, counting linked Boston entrants once. The step-1-to-step-4 differences are path-dependent accounting contributions of the formal rule, venue supply, and runner sorting, with a common upstream driver; participation selection is examined separately below by comparing take-up among actual effective qualifiers. Because the baseline is random assignment itself, the sorting residual captures the total deviation from random, not just its growth since the base period. The chain uses observed course-adjusted times (not estimated runner effects), so noise in the runner fixed effects does not enter the simulated qualification rates; noise in $\hat{\delta}$ enters through neutralization and is bounded by the index band of Section 3.7.

Three patterns stand out (Table 2 reports focal years; Appendix Figures D6 and D7 plot the full chains by cohort). Under random assignment to the frozen menu, misallocation is modest and roughly flat—a common threshold over a fixed venue menu is imperfect but stable. The direct rule effect contributes about 2.2 percentage points by the 2019 cohort. The venue-supply channel—applying the same random assignment to the actual, expanding course menu—contributes 5.0 points; runner sorting beyond random assignment adds 7.7 points. The two endogenous market channels together dominate the formal threshold change; Sections 4 and 4.4 documented each.

The random-assignment baseline is deliberately neutral, and it is not innocuous: people run their local race, so part of the step-4 deviation from random assignment is geography rather than incentive. Two geography-respecting alternatives—assigning each cohort's runners to the venue distribution of its inframarginal runners, and a gravity baseline with distance decay calibrated to observed travel—*raise* the sorting channel rather than explain it away, because full-field venue shares already contain the marginal runners' own sorting. Across every baseline and ordering of the chain, sorting contributes 5.5–7.8 points and supply 3.2–5.1: candidate sorting is at least as large as venue supply, and venue supply is never negligible (Appendix Table D5 reports the full grid).

Table 2: Misallocation accounting at focal years. Steps s_1–s_4 form the random-assignment chain. The separately labeled linked-entrant column reports the direct Boston-field estimand; it is not an additional additive channel.

	Rates (%)					
Year	s_1 Random	s_2 +Threshold	s_3 +Menu	s_4 +Sorting	Linked entrants	Coverage
2004	6.2	6.2	6.1	9.4	14.5	49.5
2010	6.2	6.2	7.1	11.5	16.5	51.1
2015	5.8	7.1	10.3	14.9	19.6	50.6
2019	5.3	7.5	12.5	20.2	28.5	44.6
2024	3.9	6.5	11.3	15.7	26.4	35.9
2025	4.1	7.2	13.4	19.4	29.1	40.0
2026	4.2	7.9	13.3	19.4	29.7	37.2

Notes: Misallocation rates in percent, US road-course qualifying universe, course effects in the Boston-state numeraire. Step 1 assigns each cohort's runners randomly to the frozen 2005–07 course menu at the 2005 effective cutoff; steps 2–4 add the cohort's actual cutoff, menu, and assignment. The linked-entrant column counts Boston entrants with an observed effective qualifier once and requires no Boston-equivalent effective qualifier anywhere in the observed window. Cancelled Boston 2020 and the atypical 2021 field are omitted from the participant line.

5.4 From qualification to participation: take-up

Boston admits a fixed field from qualified applicants, so observed appearance combines application, acceptance, attendance, and successful linkage. I measure take-up only among performances clearing the effective admission cutoff, not among everyone who merely clears the posted application standard. Across cohorts the resulting observed Boston-appearance rate is about 31.4%. Entity-resolution failure can still deflate the level and can bias cross-course comparisons if match rates differ by course. Restricting to linkage-homogeneous US-resident effective qualifiers (81% carry a validated US residence), the fast-course premium falls from 9.7 to 6.7 percentage points and the qualifying-venue multiple over the majors from 2.6 to 1.9. An external check against BAA-published qualifying-race counts for Boston

2025 shows similar recovery across course types (84% at Chicago versus 80% at CIM). The 2026 comparison is retained as a flagged sensitivity rather than the trusted endpoint because the broader course-level recovery spread is larger.

Take-up is consistently higher at faster courses among runners who clear the effective cutoff, consistent with revealed BQ intent. The direct participant line therefore complements, rather than forms a fifth additive step in, the random-assignment chain. Appendix Figure A5 compares the fast-course share among effective qualifiers with the share among linked Boston entrants; the remaining gap combines genuine participation selection and any residual differential linkage.

5.5 Course-specific handicaps and the counterfactual field

Venue-adjusted standards address the screening distortion and, by removing the qualifying advantage of fast venues, would weaken both sides of the market distortion in equilibrium. The exercises here are partial equilibrium—course choices held fixed—so they measure the immediate reallocation, with an equilibrium-response bound at the end.

The natural rule adjusts each course's qualifying times by its estimated speed: proportional handicaps, applied uniformly across age and gender groups. Most courses are in fact slower than Boston, and their runners are penalized by the common standard just as surely as fast-course runners are subsidized; a venue-adjusted standard corrects both directions, while public discussion fixates on the downhill extreme and misses half the reallocation.

Holding the field size fixed and ranking all candidates by behavior-adjusted margin (the index constructed in Section 3.5; the full suite of corrections bounds the result), 11,045 runners—13.0% of the 84,925-runner 2018–19 qualifying field (bootstrap 95% CI 12.8–13.2%)—would be displaced and replaced one-for-one. The displaced come overwhelmingly from fast courses (89.2%); the replacements come from courses at Boston's pace or slower (Figure 6, Panel A). Displacement is 9.7–14.0% across the full index suite (Panel B; Appendix Table D6 reports every index and rule), so the conclusion survives every correction of Section 3.

Two further checks address estimation error, because ranking by $T_i e^{-\hat{\delta}_{r(i)}}$ converts noise in $\hat{\delta}$ into mechanical displacement. First, a leave-out design: re-estimating the index with 2018–19 excluded—so that no qualifying performance being ranked contributes to the course effects that rank it—moves displacement from 14.5% to 13.2% on the identical field (course-level correlation 0.991 between the two indices). Second, a pure-noise placebo: simulating course effects $\delta_c \sim N(0, \widehat{se}_c^2)$ under zero true venue heterogeneity generates displacement of only 1.7% (95% range 1.3–2.0%)—an order of magnitude below the headline. Nor is displacement knife-edge churn: the median displaced runner was 3.0 minutes inside the posted standard, only 19% were within one minute, and replacements sat a median 5.5 minutes outside it, with course-neutralized times 11.9 minutes faster than the runners they displace.

The venue-adjusted field would also be faster at the prize venue. Using each runner's permanent ability (runner effect plus age–gender profile) to predict Boston-day performance, replacement qualifiers are predicted to run about 7.9 minutes faster at Boston than the runners they displace (Appendix Figure A6)—the displaced qualify on venue, not on ability.

6 What Should Boston Do?

Boston needs a rule that is accurate enough to improve comparability, transparent enough to administer, and difficult for runners or venues to arbitrage. This section compares four choices: retain a common cutoff, apply the BAA's one-sided net-drop penalties, use a simple symmetric physical formula, or handicap qualifying times with the behavior-adjusted course effects. The comparison is conducted on U.S. road marathons, the relevant qualifying market. Trail and mountain races remain useful for understanding measurement but should not determine the road-marathon policy. Figure 6 puts the four rules on one fixed field. The BAA's announced net-drop penalties replace only 2.1% of the current qualifier field. A simple symmetric elevation formula replaces 7.5%, while the behavior-adjusted course-effect rule replaces 13.0%. The gap is not a claim that more turnover is intrinsically better; it measures how much of today's venue advantage each rule actually reaches. Any change also needs a transition rule: the displaced would be runners who followed published rules in good faith, so a new standard would sensibly apply prospectively, to qualifying windows announced in advance—as the BAA's own 2027 rule does.

6.1 The case against the net-drop proxy

The BAA's 2027 rule penalizes net elevation drop alone. The rule has a real case: it is maximally simple, it could be published two years in advance, and it directly deters the extreme-downhill product category. As a measurement device, however, it is weak. Among profiled U.S. road courses, net drop explains only 0.07 of course-speed variation out of sample, compared with 0.33 for gross climbing and 0.43 for a symmetric elevation-plus-weather formula—and even the fullest physical formula explains only 0.12 of the variation weighted by where runners actually race. A one-sided net-drop rule therefore misses on both sides: fast-flat qualifying products escape unpenalized, while some large-net-drop courses with heavy climbing are overcorrected (Section 4.4.4). Any narrow proxy also creates a new target, inviting re-sorting toward whatever favorable venues the formula leaves unpenalized.

6.2 Rule classes and the next round

The four practical choices create different next-round margins. Keeping the *common cutoff* is maximally transparent but preserves the distortion. A *cardinal handicap* based on this paper's Empirical-Bayes course index best equalizes measured venue speed, but inherits estimation error and must be re-estimated as courses change. A *symmetric physical formula* using gross climbing and descent, maximum elevation, and typical weather is easier to publish and can score new courses, but its out-of-sample fit is materially weaker. The BAA's *one-sided net-drop proxy* is simplest and least accurate. An ordinal top-X% within-venue rule is a conceptually useful fifth benchmark—ranking within a race cancels any common venue effect—but creates a severe field-shopping margin and favors small, manipulable races. The choice is therefore among induced behaviors, not only approximations to δ_c. Corrections must also choose which components to neutralize—venue physics yes, effort no—and any published correction becomes the next target.

Reweighting near-BQ runners to the inframarginal course distribution—the no-incentive counterfactual—removes about half of the near-BQ fast-course excess and lowers the misallocation rate among apparent qualifiers by roughly five percentage points in 2018–19; equilibrium exit of marginal demand would additionally discourage fast-course entry, so the fixed-choice numbers above are a lower bound on the policy's full effect. The bound's assumption is that inframarginal runners' venue distribution is not itself fast-tilted for non-BQ reasons. Finally, any published handicap opens the next round: runners can chase courses with imprecise or stale handicaps, and suppliers can design to the formula. How often the index is re-estimated, whether handicaps apply ex ante or ex post, and how new courses are scored will determine how much of the screening distortion survives. A static handicap reduces today's misallocation; it does not by itself close the loop.

7 Conclusion

A common qualifying threshold applied across heterogeneous, manipulable evaluation environments produces two linked distortions, and the observed venue value-added a naive fixed-effects index reports is a composite of physical venue speed and the behavior the threshold induces. This paper builds the index, tests its assumptions in the open, documents the market the threshold created, prices the misallocation, and corrects the measure for the behavior it absorbs.

The screening distortion is large. Among linked Boston entrants with an observed effective qualifier, 29.1% in the trusted 2025 cohort have no observed performance that would still clear the effective cutoff on a Boston-equivalent course—roughly double the 2004 share—and a venue-adjusted standard would displace about 13.0% of recent Boston qualifiers, replacing them with runners predicted to be faster at Boston itself. The market distortion is two-sided: near-threshold runners sort to fast venues, with the response growing after announced tightenings, and purpose-built qualifying products entered until the supply channel contributed misallocation of the same order as candidate sorting under every counterfactual baseline. The screening rule does not merely rank runners imperfectly; by the end of the sample period it has reshaped the market in which qualifying performances are produced.

These distortions cannot be repaired by targeting a few extreme courses. In the policy-relevant U.S.-road sample, gross climbing and typical weather predict speed better than net drop, but even the full physical formula leaves most finisher-weighted variation unexplained out of sample. Index-based venue adjustment is therefore the most accurate mature-course rule; a symmetric physical formula is a transparent fallback for new courses; and every published correction opens a next round.

For measurement practice, the marathon market is a laboratory in which the full Goodhart loop around a fixed-effects performance measure is visible end to end, with ground truth on venue physics and a retest at the prize venue. The tools generalize: when a fixed-effect measure becomes the prize, exogenous mobility fails by design, predictably, and locally—signed toward favorable venues, concentrated near the threshold, intensifying with each tightening—which makes it detectable through threshold-distance interactions, moving-threshold and placebo-threshold tests, and directional mover asymmetries; strategic shares scale the correction; and audits of candidate gaming alone miss the supply side, which here grew to first order within a decade.

One caveat raises the credibility of the rest: detection in this setting is powered by a threshold that is public, sharp, and moves on announced dates. Where operative thresholds are murkier—teacher evaluation, hospital report cards—the toolkit's power is correspondingly an upper bound. What carries over is the logic: measure the components, exploit the institution's own discontinuities, and audit both sides of the market.

The broader lesson is that common thresholds are not neutral when performance is produced in environments candidates choose and suppliers build. In admissions, accountability, certification, and ratings, the same loop operates with the environments harder to see; here, every link of the loop—rule, candidates, venues, and measure—is observed.

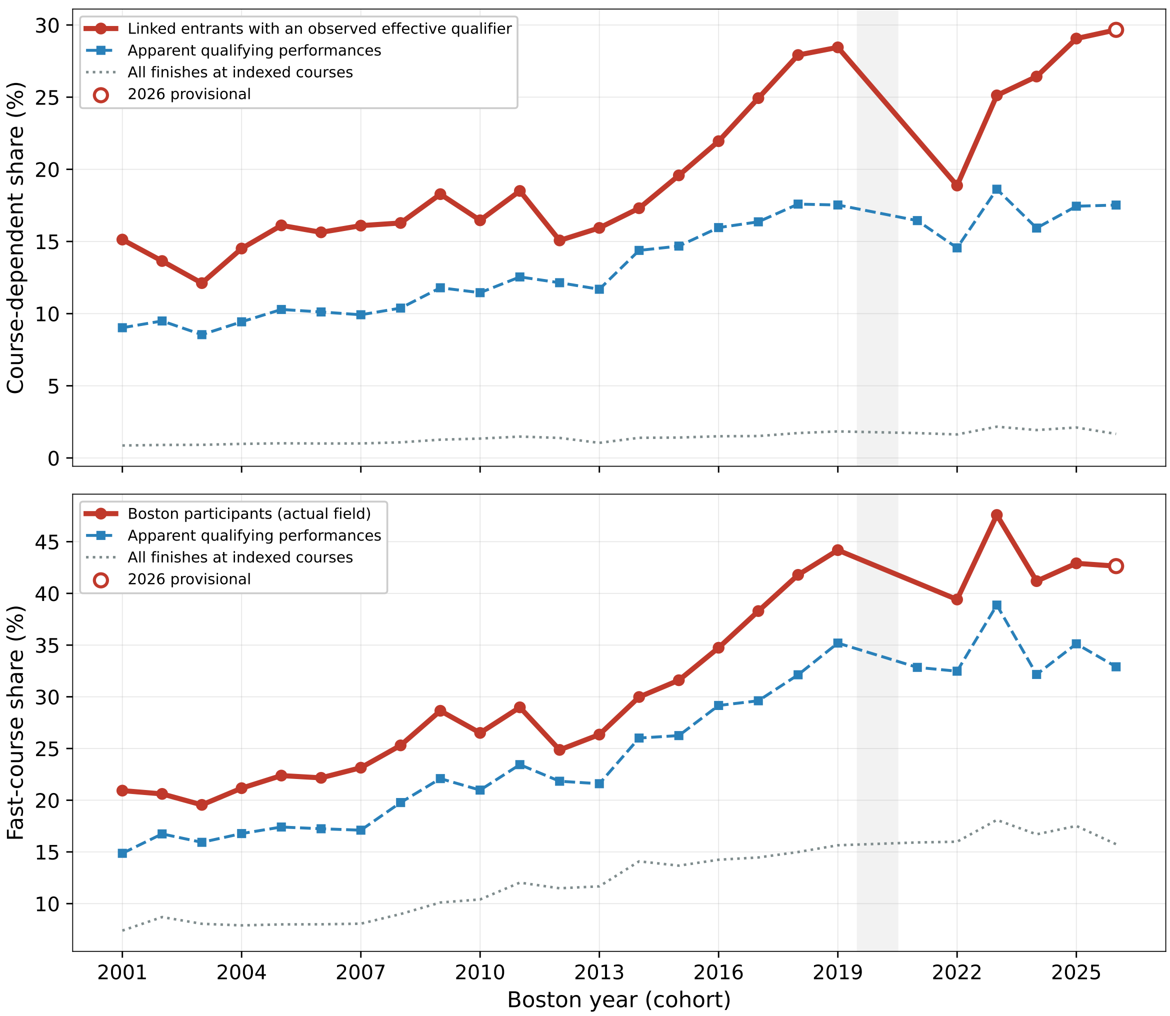
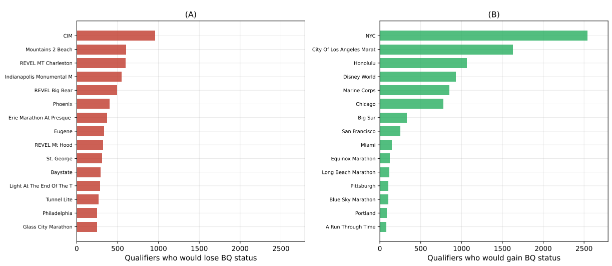
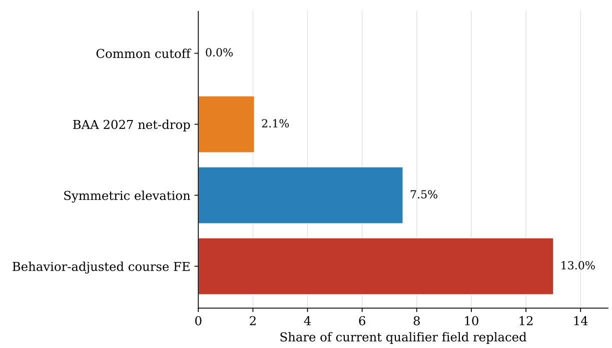


Figure 5: Course-dependent qualification has risen. Panel A plots the share of each population whose qualification depends on its course: linked, classifiable Boston entrants at the effective admission cutoff (solid), apparent qualifying performances at the posted standard (dashed), and all finishes at indexed courses (dotted). Panel B plots the corresponding fast-course shares for the same three populations; there the solid series is the actual Boston field. The shaded band marks the cancelled 2020 race; Boston 2025 is the latest trusted endpoint and 2026 is shown as provisional (open marker).



(a) Courses losing / gaining qualifiers



(b) Displacement by index / rule

Figure 6: What alternative Boston rules would change. Panel A shows which courses lose and gain qualifiers under a venue adjustment. Panel B compares fixed-field displacement across the common cutoff, the BAA net-drop penalties, a symmetric elevation formula, and the behavior-adjusted course effects.

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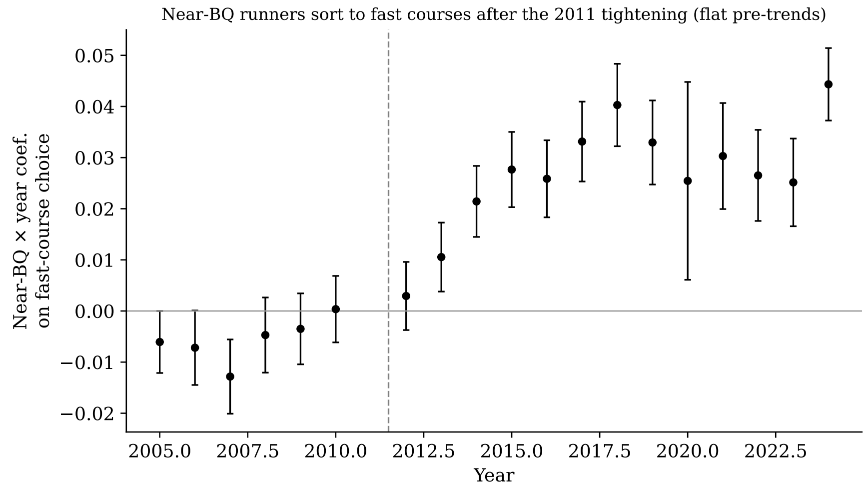


Figure A1: Event study of near-BQ runners' differential fast-course choice, relative to 2011 (95% CIs). Flat pre-trends; sustained rise after the February 2011 announcement (the first post-announcement full qualifying year is 2012).

Appendix A Additional Behavioral, Supply, and Policy Evidence

The following exhibits support the six-figure main-text argument and provide the additional timing, supply, take-up, and correction detail.

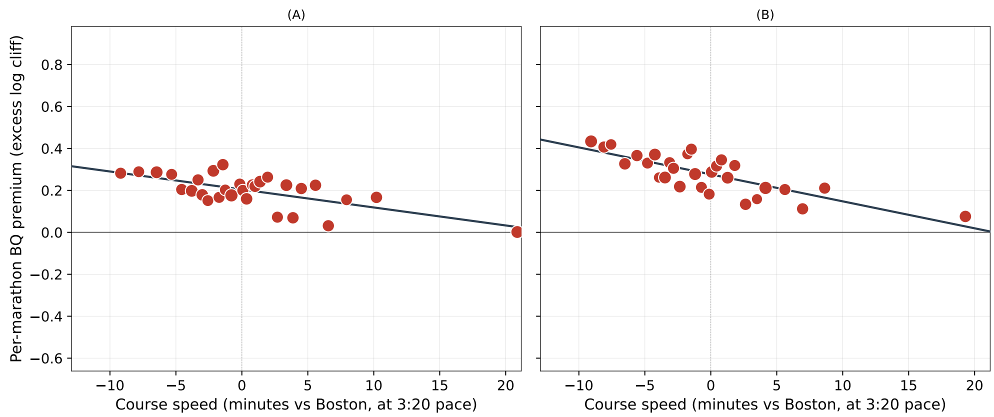


Figure A2: Per-course BQ-bunching premium by course speed, before and after the 2011 tightening: the course-speed gradient in BQ targeting steepens from -0.0088 to -0.0131 per minute of course speed.

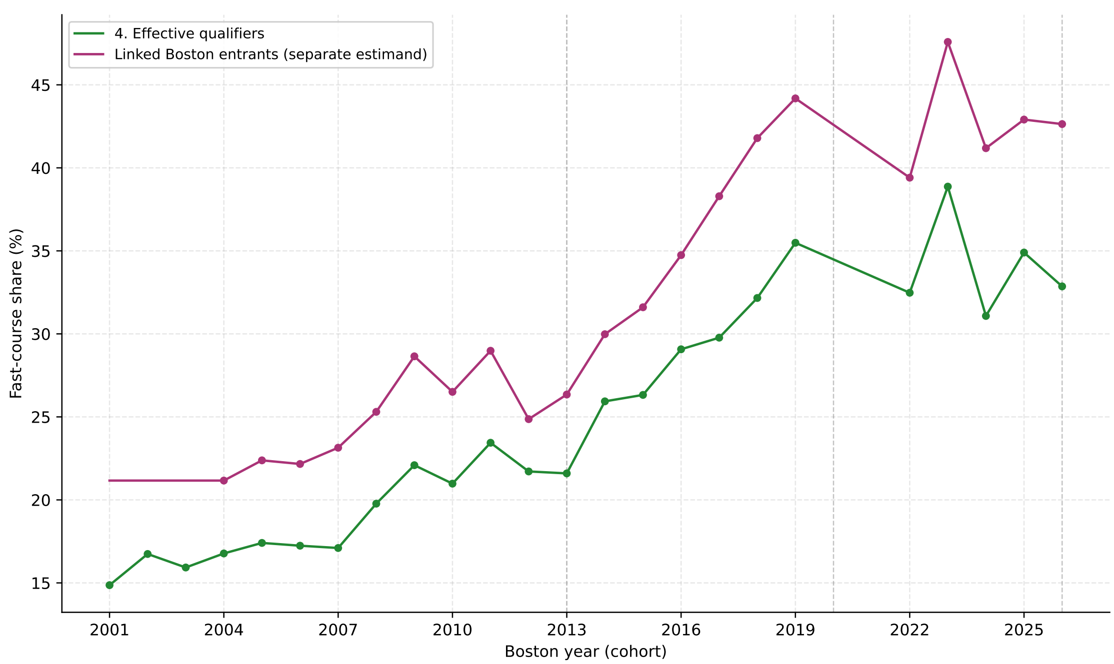


Figure A5: Fast-course exposure among effective qualifiers and among linked Boston entrants with an observed effective qualifier. Each participant is counted once using the fastest observed performance clearing the effective cutoff. COVID-era cohorts omitted.

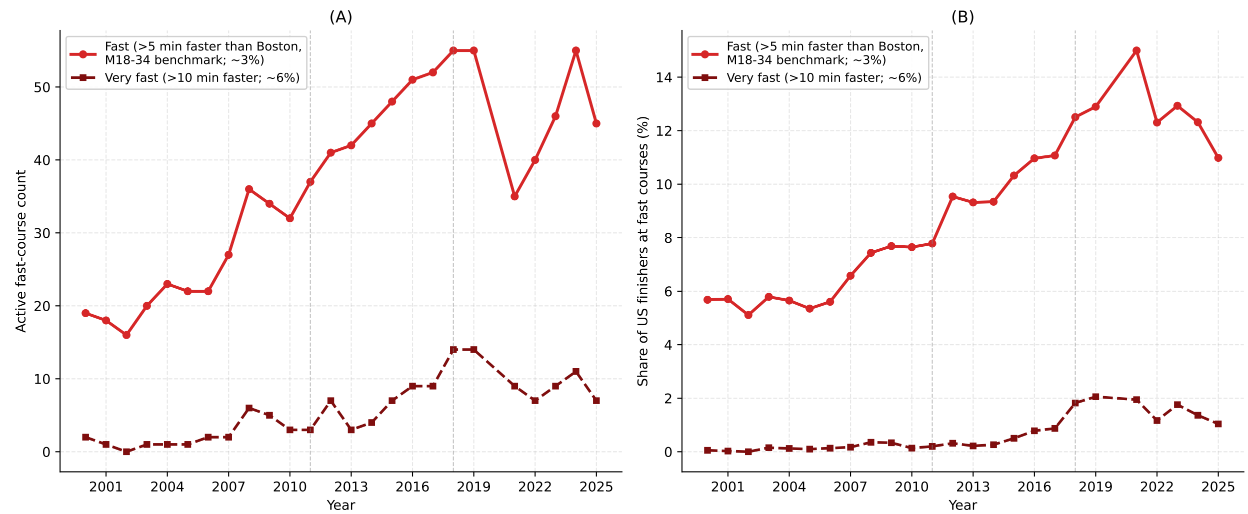


Figure A3: Growth of the fast-course market, 2000–2025. Panel A: count of US courses at least five minutes (fast) and ten minutes (very fast) faster than Boston at the M18–34 standard. Panel B: share of US finishers at such courses. Dashed lines mark the 2011 and 2018 announcements.

Appendix B Measurement Details: AKM Specification, Shrinkage, and Course Drivers

This appendix gives the AKM specification and Empirical Bayes shrinkage behind the course-speed index of Section 3, an algebraic invariance result for the displacement accounting, the within-runner additive-separability validation behind Section 3.3.1, the observable predictors of course speed, and the contamination-diagnostics exhibits behind Section 3.

B.1 AKM Estimator

I estimate the following specification:

$$\ln T_{irt} = \alpha_i + \delta_r + \gamma_t + f(\text{age}_{it}, \text{male}_i) + g(W_{rt}) + \varepsilon_{irt} \tag{5}$$

where T_{irt} is the finish time (in seconds) of runner i in race r in year t, α_i is a runner fixed effect capturing persistent baseline performance, δ_r is a race fixed effect capturing

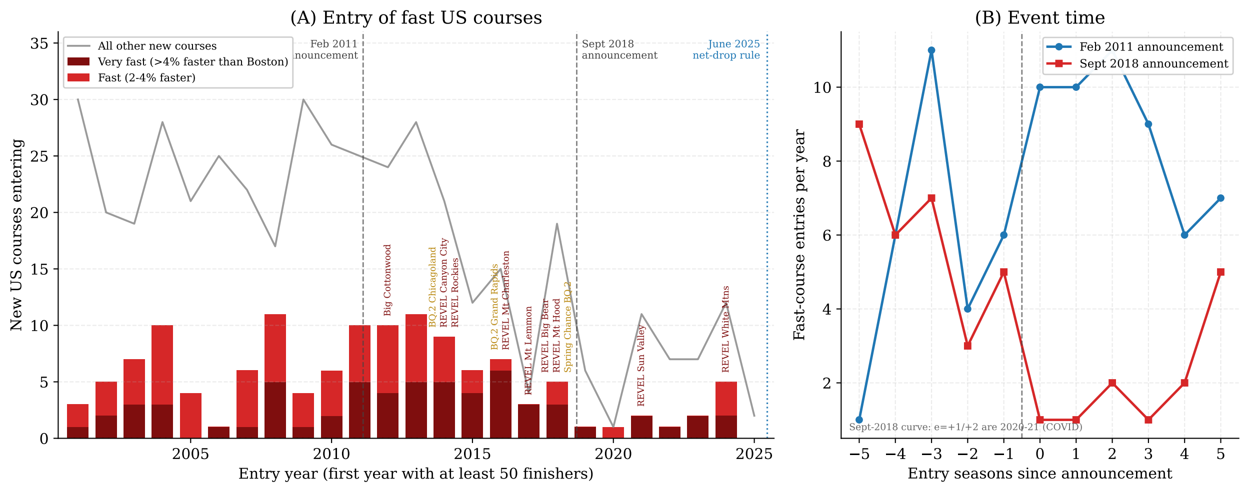


Figure A4: Entry of US road marathon courses by course-speed class, 2001–2025. Panel A: entry year and speed class, with dedicated qualifying products labeled by name; dashed lines mark the February 2011 and September 2018 announcements and the June 2025 net-drop rule. Panel B: fast-course entries per year in event time around the two announcements. Entry year is a course's first panel year with at least 50 finishers; courses already active in 2000 are left-censored incumbents and excluded.

the persistent speed of course r (including its typical climate), and γ_t is a year fixed effect capturing common trends across all races. The function $f(\cdot)$ is a quartic polynomial in age, centered at 40, fully interacted with gender. The term $g(W_{rt})$ is a quadratic in demeaned race-day temperature, $g(W_{rt}) = \beta_1 \widetilde{W}_{rt} + \beta_2 \widetilde{W}_{rt}^2$, where $\widetilde{W}_{rt}$ is the deviation of race-day morning temperature from the marathon's historical mean. Because mean temperature is absorbed by the course fixed effect δ_r, the weather controls capture only transitory race-year temperature shocks. (Humidity, wind, and precipitation add little explanatory power *on average* and are omitted from the model; they can matter at specific exposed courses, which is visible in the within-course stability diagnostic of Appendix E.)

To reduce false links in the entity resolution procedure, I define runner identities separately within broad geographic regions (U.S., Canada, UK/Ireland, Australia/New Zealand, and Other), so the same physical runner competing in both the U.S. and UK is assigned separate fixed effects in each region. The model is estimated using `reghdfe` (Correia, 2017), with standard errors clustered by regional runner.

The key assumption is additive separability: after accounting for runner ability and

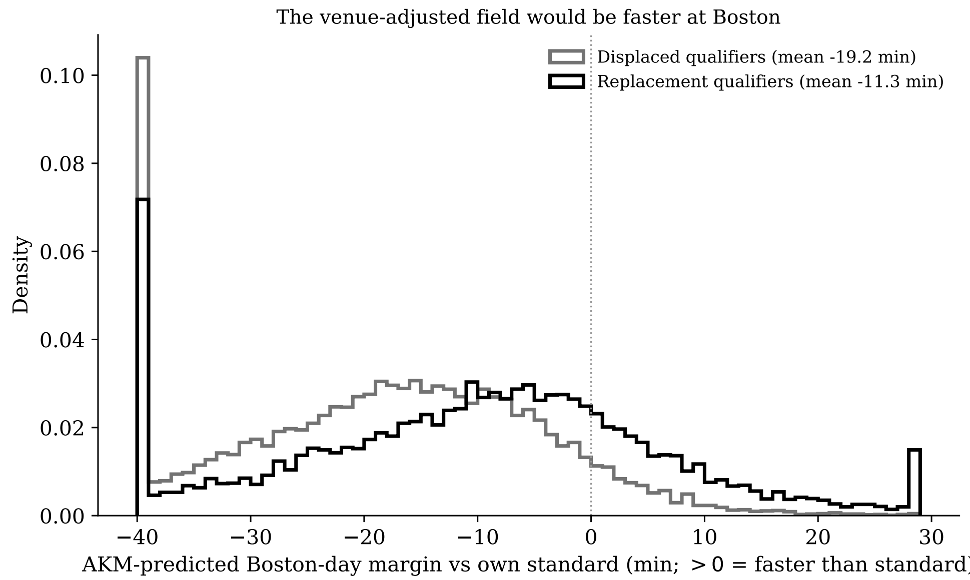


Figure A6: Predicted Boston-day performance of displaced vs. replacement qualifiers (AKM runner effects; margins relative to each runner's own standard).

observables, courses shift log finish times by a common amount. This is a plausible approximation for road marathons, where features such as elevation profile, terrain, and typical weather conditions tend to affect all runners proportionally in time rather than creating highly runner-specific match effects. Appendix E reports two diagnostic exercises in the spirit of Card, Heining and Kline (2013): within-course year-to-year stability of course effects, and a nonseparability test. Additive separability (Section 3.3.1 in the main text; regression detail in Section B.4; $\hat{\beta} \approx 0.851$ for Boston-destination and 0.933 for general cross-race pairs) supplies the mover-style cross-venue validation that is typically the third CHK diagnostic. Taken together, these exercises suggest that the additive structure is a useful approximation, though not an exact description of every runner-course combination.

The overall R^2 of equation (5) is 0.816 on 11,142,687 observations. Persistent runner differences account for the bulk of that fit; the race fixed effects contribute a modest additional share that is nevertheless the object of interest, and year effects and weather controls add a small further increment. The weather controls are significant: a 10°F warmer-than-average race day slows finish times by 2.2%, or about 4.3 minutes at a 3:20 pace.

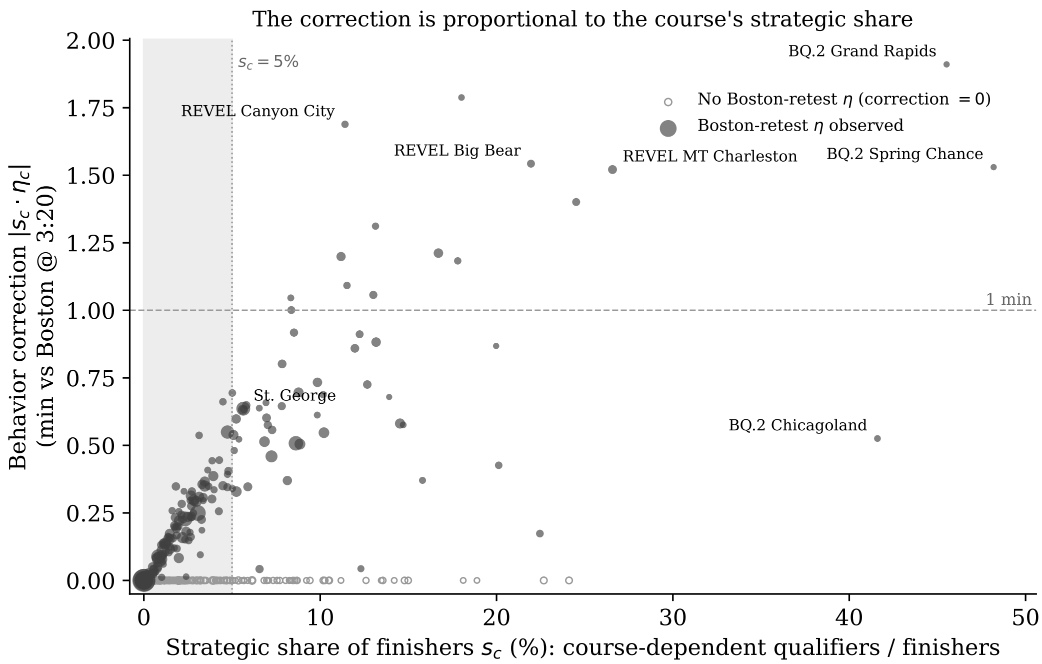


Figure A7: Behavior correction is proportional to the strategic share. Each point is a US road course: the horizontal axis is the strategic share s_c (course-dependent qualifiers as a share of finishers), the vertical axis the behavior correction in minutes at 3:20 pace, and marker size is proportional to finisher count. Courses without a measured peaking residual receive no correction by construction (open circles); shading marks $s_c < 5\%$; dedicated qualifying venues are labeled.

B.2 From Fixed Effects to a Course Speed Index

The course speed index is constructed from the estimated race fixed effects $\hat{\delta}_r$, which capture persistent differences in venue advantage after netting out runner ability, age-gender profiles, and race-year shocks. Because the model is estimated separately within each geographic region, these fixed effects are identified only up to a region-specific constant. I normalize the U.S. estimates by setting $\hat{\delta}_{\text{Boston}} = 0$ and calibrate the remaining regions using approximately 24,000 trusted bridge runners who appear in both the U.S. and another region. For each non-U.S. region, I use the median difference in runner fixed effects among bridge runners to recover a calibration constant that places all course effects on the Boston-centered U.S. scale.

I then apply Empirical Bayes shrinkage to the estimated course effects to reduce noise from sparsely connected courses. Intuitively, the procedure treats each estimated $\hat{\delta}_r$ as a noisy signal of an underlying permanent course effect and pulls imprecisely estimated values toward the overall mean, with stronger shrinkage for courses estimated with greater variance (Morris, 1983; Andrews et al., 2008). Letting $\widehat{V}_r$ denote the estimated sampling variance of $\hat{\delta}_r$ and $\hat{\tau}^2$ the cross-course variance of the underlying effects, the shrinkage factor is

$$w_r = \frac{\hat{\tau}^2}{\hat{\tau}^2 + \widehat{V}_r},$$

so the shrunk estimate is

$$\hat{\delta}_r^{shr} = w_r \hat{\delta}_r.$$

Courses with little information are therefore pulled more strongly toward zero, while well-measured courses are left largely unchanged. I retain courses with at least 50 AKM observations after shrinkage; the final index covers 1057 courses.

Because the model is estimated in logs, the course effect $\hat{\delta}_r^{shr}$ is a proportional measure: a course with $\hat{\delta}_r^{shr} = -0.05$ is approximately 5% faster than Boston for all runners, regardless of pace. This proportional property means a given percentage advantage translates to more minutes for slower runners. For a runner on pace to finish in 3:00, a 5% course advantage is worth 9 minutes; for a 3:20 runner, 10 minutes; for a 4:00 runner, 12 minutes.

Throughout the paper, I report course effects in percentage terms and provide minute equivalents at two reference paces: 3:00, the 2025 M18–34 qualifying standard, and 3:20, which approximates the pooled qualifier finish time (median 3:19:21) across 968,714 qualifier-races in the sample. I define a "fast" course as one that is at least 2% faster than Boston—equivalent to about 3.6 minutes at a 3:00 pace or 4 minutes at a 3:20 pace—which captures 250 courses (24% of the index). I define a "very fast" course as one at least 4% faster (110 courses, 10%). I define the BQ handicap as the percentage by which the qualifying standard should be changed to equalize course difficulty; because it is proportional, a single handicap applies to all age and gender groups.

B.3 An Invariance Result

One algebraic fact disciplines the displacement accounting. The fixed-field counterfactual in Section 5 ranks all candidates by venue-adjusted time $T_i e^{-\hat{\delta}_{r(i)}}$ and admits a fixed number. Adding any constant k to every course effect rescales every adjusted time by e^{-k} and leaves the ranking—hence the admitted set, hence the displacement share—unchanged. The ranking invariance has three consequences. First, the choice of numeraire cannot affect the displacement headline: whether "Boston = 0" means Boston in race-day state or Boston in qualifying-attempt state is irrelevant. Second, any economy-wide component of effort or peaking absorbed into the index is irrelevant. Third, *only cross-course differential contamination matters*—which is precisely what the diagnostics of Section 3 measure and what the corrected-index suite bounds. The misallocation *rate* against the posted standard is not invariant to k; that series is therefore always presented with its Boston-state numeraire stated, and the cross-index band serves as its robustness check.

B.4 Validation: Cross-Race Additive Separability and the Boston Retest

The estimated course effects are useful only if they capture real, additively separable differences in venue speed. The cleanest test is whether the index is predictive of subsequent Boston race performance. The regression model below assesses this prediction. I then generalize to the full set of cross-race pairs as a broader check.

The core specification, applied throughout this subsection, is:

$$\ln T_{i,\text{dest}} - \ln T_{i,\text{orig}} = \beta \cdot (\hat{\delta}_{\text{dest}} - \hat{\delta}_{\text{orig}}) + \mathbf{X}_i' \boldsymbol{\phi} + \varepsilon_i \tag{6}$$

where $\hat{\delta}_{\text{dest}} - \hat{\delta}_{\text{orig}}$ is the difference in estimated course effects between the destination and origin races and $\mathbf{X}_i$ collects a quadratic in age and gender. By differencing finish times within runner, this specification absorbs time-invariant runner ability. Perfect additive separability implies $\beta = 1$: moving to a course that is 1 log point faster should improve finish time by 1 log point on average. Standard errors are clustered by runner.

B.4.1 Boston as a Standardized Retest

Because Boston is the normalization point of the course index, $\hat{\delta}_{\text{Boston}} = 0$ by construction. I construct the retest sample by taking every runner-year in which the runner ran Boston and pairing it with each of that runner's marathons in the 18 months before Boston at a different course, yielding 690,058 pairs. On average, runners are slower at Boston than at their prior race, and faster qualifying courses predict larger Boston slowdowns (Figure 2, Panel A).

Table B1, column 2, reports estimates of equation (6) on the Boston-destination subsample. The estimated additive separability coefficient is $\hat{\beta} = 0.851$: for each 1% increase in a prior marathon's speed, a runner is predicted to run Boston about 1% slower. On a raw basis, runners are on average slower at Boston than at their prior race; most of this is explained by course-FE additive separability, leaving a positive constant that captures residual Boston-specific factors such as fatigue, travel, race-day conditions, and the order/peaking behavior measured directly in Section 3.4. (Figure 2 in the main text plots both retest panels.)

B.4.2 General Cross-Race Additive Separability

To show that additive separability is not specific to Boston, I generalize the pair construction to all destinations. For every runner-race observation in the dataset, I pair it with each prior marathon by the same runner in the 18 months before at a different course.[4] This yields 10,934,525 pairs. Figure 2, Panel B, shows the binscatter; Table B1, column 1, reports $\hat{\beta} = 0.933$. Column 4 estimates an interacted model that allows the slope to differ between Boston and non-Boston destinations; both the Boston-destination ($\hat{\beta} = 0.851$) and non-Boston-destination ($\hat{\beta} = 0.934$) coefficients closely approach the 1.0 benchmark of perfect

[4] The same-course restriction excludes pairs like Boston 2019 / Boston 2020, which would contribute a degenerate zero FE-difference observation; the runner still enters the analysis through cross-course pairs (e.g., Boston 2019 / NYC 2018 and Boston 2020 / NYC 2018).

additive separability.

Table B1: Cross-race additive separability of course fixed effects

	(1) All pairs	(2) Boston dest.	(3) Non-Boston	(4) Interacted
FE difference	0.933***	0.851***	0.934***	
	(0.002)	(0.009)	(0.002)	
FE diff × Boston				0.852***
				(0.009)
FE diff × non-Boston				0.934***
				(0.002)
Boston-dest indicator				0.042***
				(0.000)
Age controls	✓	✓	✓	✓
N	10,934,525	690,058	10,244,467	10,934,525
R^2	0.156	0.070	0.161	0.162

Notes: Dependent variable is the log time gap, $\ln T_{\text{dest}} - \ln T_{\text{origin}}$. "FE difference" is the course fixed effect at the destination minus the origin. Each runner's destination race is paired with every prior marathon at a different course within 18 months; pair-weighted WLS so prolific runners do not dominate; age and gender controls. Column (4) interacts the FE difference with a Boston-destination indicator. Standard errors clustered by runner in parentheses. $^{***}p < 0.001$.

Table B2: Additive-separability slope by pair stratum: name rarity and entity-resolution link provenance

	All destinations			Boston destination		
Pair stratum	$\hat{\beta}$	(SE)	Pairs	$\hat{\beta}$	(SE)	Pairs
All pairs	0.933	(0.001)	10,934,525	0.851	(0.004)	690,058
Rare names ($\leq$ 2 people per name)	0.895	(0.001)	6,963,864	0.774	(0.004)	482,583
Medium (3–10 people)	1.006	(0.002)	2,939,352	0.961	(0.009)	152,046
Common ($>$ 10 people)	1.162	(0.003)	1,031,309	1.357	(0.019)	55,429
Excluding attached yob-less links	0.936	(0.001)	5,272,574	0.857	(0.005)	441,583
Excluding residence-merge links	0.915	(0.001)	4,861,866	0.810	(0.005)	387,801
Medium, all low-confidence links excluded	0.981	(0.003)	710,472	0.928	(0.014)	62,513
Rare, all low-confidence links excluded	0.876	(0.001)	2,098,419	0.741	(0.006)	208,080

Notes: WLS estimates of equation (6) within pair strata (pair-weighted, age and gender controls). Tiers classify the runner's (first name, last name, gender) cell by the number of distinct people carrying it, measured as distinct birth-year clusters. "Attached yob-less links" are runner ids that mix rows with and without a birth interval (the entity resolution's NA-attach step); "residence-merge links" are ids consolidated by the modal-residence re-merge. Slopes above one among common names indicate occasional false merges—a spurious mover pairing a faster person with a faster course—while slopes into Boston sitting below slopes elsewhere indicate origin-race peaking (Section 3.4). Excluding the entity-resolution merge steps leaves the slopes essentially unchanged, so the deviations are not artifacts of the linkage design.

B.5 What Drives Course Speed?

Course fixed effects summarize everything that makes a marathon systematically faster or slower after accounting for runner ability, age, year, and realized weather. That breadth is their advantage, but it also makes the index a "black box." To ask how much transparent course information can recover, I use the same policy-relevant universe as the main analysis: 439 profiled U.S. road courses with net and gross elevation, maximum elevation, and typical weather. I estimate

$$\widehat{\delta}_r = \mathbf{X}_r'\boldsymbol{\beta} + u_r, \tag{7}$$

where $\widehat{\delta}_r$ is the EB-shrunk log course effect and $\mathbf{X}_r$ contains observable course characteristics.

Table B3 separates descriptive fit from genuine prediction. All columns use exactly the same courses. Net drop alone predicts little; gross climbing is substantially more informative, and combining net and gross elevation improves fit further. Adding maximum elevation and typical temperature, wind, and precipitation produces an in-sample R^2 of 0.57, but only 0.47 when predicting held-out courses. Because the AKM target is approximately 0.96 reliable, sampling error can raise that predictive fit only to 0.49. The remaining error therefore primarily reflects omitted or coarsely measured features—grade sequencing, turns and loops, route vintages, crowding, exposure, and race operations—not an imprecisely estimated target.

Table B3: Observable predictors of course speed on U.S. road courses

	(1) Net drop	(2) Gross gain	(3) Net + gross	(4) Physical + weather
Courses	439	439	439	439
In-sample R^2	0.08	0.41	0.47	0.57
10-fold cross-validated R^2	0.07	0.33	0.39	0.47

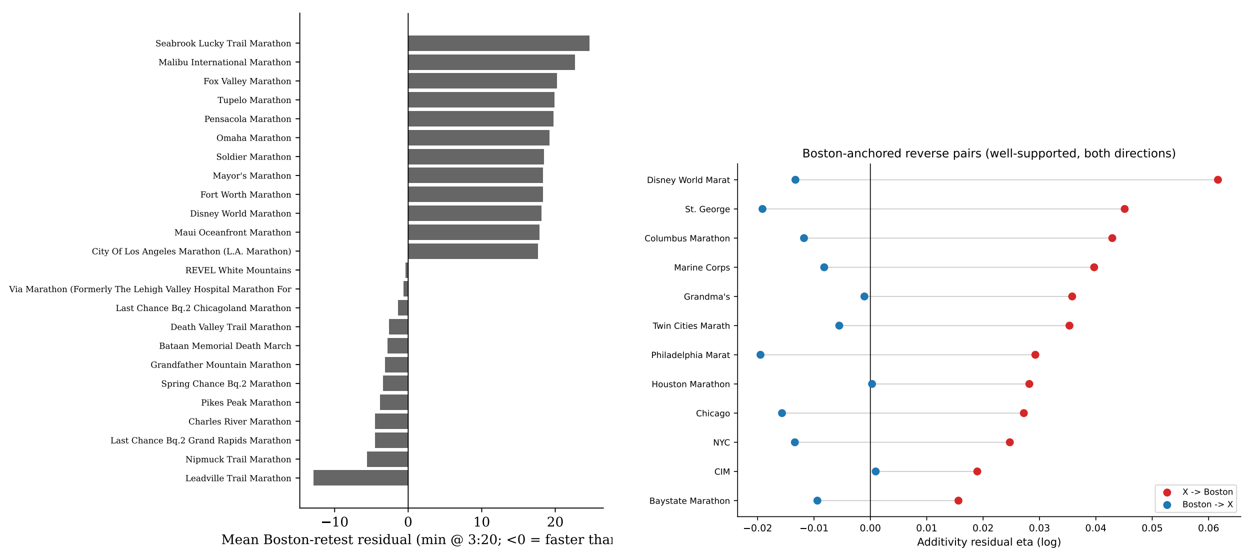
Notes: The dependent variable is the EB-shrunk AKM log course effect, normalized to zero at Boston; positive values indicate slower courses. Estimates are course weighted and use an identical sample of profiled U.S. road courses in every column. Column (4) includes net drop, gross gain, maximum elevation, typical temperature and its square, wind, and precipitation. Cross-validation randomly assigns entire courses to ten folds and predicts each fold using the other nine. Approximate repeat-sample reliability of the dependent variable is 0.96; the corresponding attenuation-adjusted column-(4) cross-validated R^2 is 0.49.

B.6 Contamination Diagnostics: Additional Exhibits

This subsection collects the exhibits behind Section 3: the destination-numeraire retest slopes, the ordered edge/peaking diagnostics, the strategic-share and behavior-adjusted-index detail, the construction of the routine-attempt and reverse-pair estimators, and the implementable conservative and observable-based variants.

Table B4: Boston retest by destination numeraire: same-runner pairs into each destination; slope of the realized log-time change on the predicted course difference. SEs clustered by runner.

Destination race	Slope $\hat{\beta}$	SE	Pairs
Boston	0.851	(0.009)	690,058
Chicago	0.988	(0.035)	11,852
New York City	0.998	(0.038)	2,438
Marine Corps	0.826	(0.039)	62,607
Philadelphia	0.824	(0.046)	19,941
All destinations	0.933	(0.002)	10,934,525



(a) Boston-retest residual by US origin course (b) Direction matters (reverse-pair asymmetry)

Figure B1: Runners peak at the qualifier, not at Boston: the flattering performance attaches to the race run before Boston, and the residual reverses with direction.

Table B5: Ordered edge/order decomposition of the additivity residual η (minutes at 3:20 pace). Columns add controls progressively; SEs clustered by runner (two-way runner $\times$ course SEs reported in the underlying output).

	(1) Order	(2) +Near-BQ	(3) +Goal/qual.
Destination is Boston	7.930	7.396	2.907
Origin is Boston	-2.134	-3.325	-6.371
Near BQ at origin		9.869	5.419
Dest. Boston $\times$ near BQ at origin		-6.062	-1.591
Qualified at origin			12.490
Dest. Boston $\times$ qualified at origin			-3.245
Major-city destination			0.296
Temperature deviation difference (°F)	0.344	0.340	0.333

Table B6: Strategic share and the behavior-adjusted index for the fastest courses. η is the Boston-retest peaking residual; "Adjusted" is equation (4).

	Naive	Strategic share (%)		Peaking η	Adjusted
Course	(min)	Finishers	Qualifiers	(min)	(min)
REVEL MT Charleston	-17.1	26.6	77.6	6.3	-15.6
Spring Chance BQ.2 Marathon	-16.6	48.2	90.0	-3.4	-18.2
Tunnel Lite Marathon	-15.6	22.5	63.5	0.8	-15.4
Last Chance BQ.2 Chicagoland Marathon	-15.3	41.6	83.9	-1.4	-15.8
REVEL Big Bear and Half Marathon	-15.1	22.0	66.5	7.7	-13.6
Last Chance BQ.2 Grand Rapids Marathon	-14.5	45.5	87.8	-4.5	-16.4
Light At The End Of The Tunnel Marathon	-12.8	24.5	71.3	6.2	-11.4
MT. Nebo Marathon	-11.4	9.8	52.1	6.7	-10.8
Hudson Mohawk Marathon	-11.2	17.8	56.0	7.1	-10.0
Carmel Marathon	-11.2	12.0	52.0	7.7	-10.3
Loco Marathon	-10.0	14.7	58.7	4.2	-9.4
Tunnel Vision Marathon	-9.9	20.1	60.1	2.2	-9.5
REVEL Canyon City Marathon	-9.6	11.4	51.9	16.1	-7.9
Bay State Marathon	-9.6	18.0	55.9	10.6	-7.8

Routine-Attempt Estimator: Construction. A natural alternative to correcting the pooled index is to split the sample and estimate the model separately for targeting and non-targeting attempts. Targeting, however, is a runner-*race* state, not a runner type: the same person targets in some attempts and not others, so the pair, not the runner, is the right unit of a split. The routine-attempt estimator implements that split. An attempt is *routine* ex ante if the runner's geocoded home is within 100 miles of the course or the runner has run that course in a prior year: locals and repeaters run Chicago because it is their race, not as a once-peaked goal. Estimating course effects from routine–routine same-runner pairs only (a within-runner pair estimator on 1.9 million pairs) gives a routine-only index whose correlation with the naive index is 0.96 across 733 courses. Each course's *visitor premium*—naive minus routine-only, after removing the common Boston-numeraire level that the invariance result (Section B.3) makes irrelevant—measures how much of its apparent speed is visitor composition. The premium concentrates where peaking predicts: dedicated goal venues are flattered by visitor composition, ordinary majors are not, and the interquartile range across courses is -2.7 to +1.5 minutes (Figure B2, Table B7). The contamination is real, signed as predicted, nameable course by course—and bounded. The split fails only where targeting is the product itself (who runs a last-chance qualifier casually?), which is why the corrected-index suite pairs it with corrections that need no non-targeting subsample.

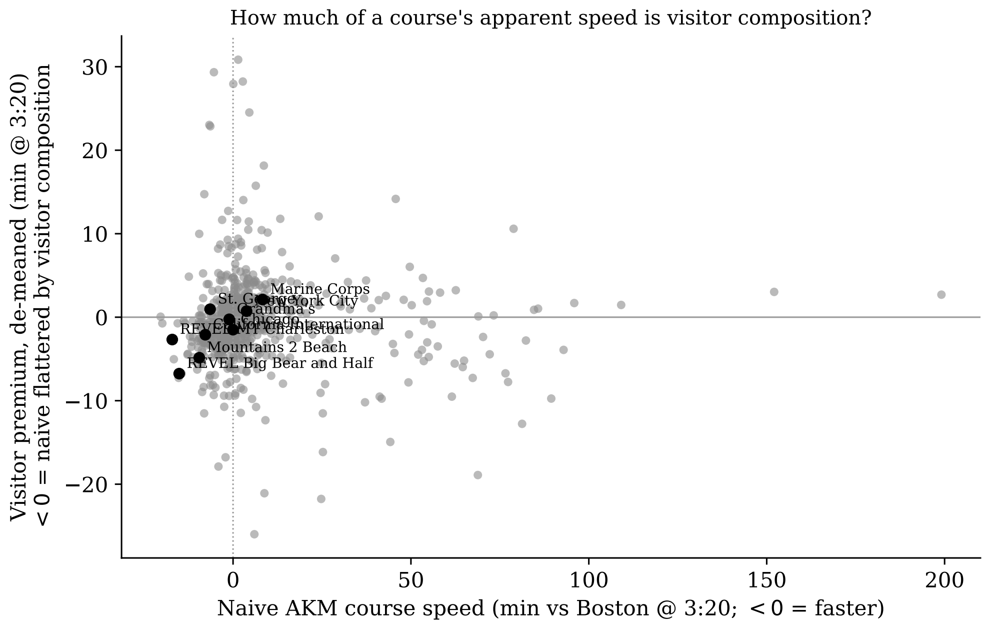


Figure B2: De-meaned visitor premium by course: negative values mean the naive index is flattered by visitor composition. Dedicated goal venues stand out; ordinary local races sit near zero.

Table B7: Routine-only course speeds and visitor premia, largest absolute de-meaned premia among well-measured courses.

Course	Naive (min)	Routine-only (min)	Visitor premium (min)	Routine share (%)	Routine pairs
Bataan Memorial Death March	78.9	56.5	10.6	27.0	4,420
Sarasota Marathon The Grouper Run	0.7	3.6	-9.1	53.9	1,551
Loonies Midnight Marathon	2.2	-12.6	9.0	30.2	1,102
Lakeshore Marathon	-1.8	0.0	-8.0	51.5	2,899
Ocala Marathon	-0.8	0.7	-7.8	50.2	1,707
Gasparilla Distance Classic	0.9	2.1	-7.4	44.0	6,014
Last Chance Bq.2 Chicagoland Marathon	-15.3	-14.4	-7.3	41.8	1,593
Honolulu Marathon	28.7	14.0	7.0	35.4	23,037
Big Island International Marathon	10.7	11.5	-7.0	30.0	2,334
REVEL Big Bear and Half Marathon	-15.1	-14.7	-6.8	45.8	7,692
Ojai 2 Ocean Marathon	-10.1	-10.1	-6.3	42.8	1,413
Five Points Of Life Marathon	-1.5	-1.7	-6.2	54.1	1,691
Deadwood-Mickelson Trail Marathon	15.9	2.9	6.1	18.9	1,789
A Run Through Time	62.3	61.3	-5.6	59.0	1,446

Reverse-Pair Estimator. Averaging the $A \to B$ and $B \to A$ directions of same-pair gaps differences out order and peaking components that are direction-specific, identifying $\delta_B - \delta_A$ under the assumption that order effects are symmetric in expectation. This estimator is the constructive use of Figure B1b and appears in the suite of Table D7.

Conservative and Observable-Based Variants. Two implementable variants discipline estimation error. A confidence-adjusted handicap shrinks each course's adjustment toward zero by k standard errors, zeroing courses statistically indistinguishable from Boston; at $k = 1.645$ displacement falls to 11.1%. An observable-based rule uses predicted speed from terrain, surface, and climate—applicable to brand new courses with no race history—and yields 9.7%; it under-adjusts the fast-flat loop courses whose speed does not come from elevation, which is also why it is not a behavior-free benchmark. A practical implementation uses the observable formula for new courses and the estimated index for established ones, updating annually.

Appendix C Data Construction

Section 2 describes the primary data sources and estimation sample. This appendix provides additional detail on the entity-resolution procedure and the selection of the race-time variable.

Table C1 summarizes the construction of the samples used later in the paper, and the sample sizes differ across rows for specific reasons that the rest of this paragraph explains. The raw linked panel is the full web-scraped dataset after entity resolution: every finish-time observation at every marathon with runner identities assigned across races. The AKM estimation sample is smaller because the model requires each runner to appear at least twice within a geographic region so that the runner fixed effect is identified, requires a known year of birth so that the age polynomial is defined, and trims the slowest 0.1% within each race-year to prevent outliers from dominating the residual variance. These three filters drop singletons, observations with missing year of birth, and extreme outliers, which together cut the observation count from 17.2 million to 11.1 million. The course count falls modestly because small races at which every runner is a singleton contribute no observations after the two-appearance requirement. The course speed index row is a course-level object, not an observation-level one; it reports only the number of courses retained for the published index, which are those with at least 50 AKM observations after shrinkage. The qualification analysis sample is the set of non-Boston observations at indexed courses in years with a known BQ standard. It is drawn from the full linked panel rather than the AKM sample, so singletons reenter (their BQ margin is defined without a runner fixed effect) and the observation and runner counts rise correspondingly. The cross-race additive separability pairs are a pair-level object: for each runner-race in the linked panel, I form a pair with every prior marathon by the same runner at a different course within 18 months. The runner count is lower than in the AKM sample because only runners with at least two appearances within an 18-month window can generate a pair. The Boston-destination pairs are the subset whose destination is Boston, which defines the standardized retest because the index is normalized so that $\hat{\delta}_{\text{Boston}} = 0$.

Table C1: Sample construction

Sample	Observations	Runners	Courses
Raw linked panel	17,206,046	8,124,265	1,172
AKM estimation sample	11,142,687	2,631,055	1,162
Course speed index	—	—	1057
Qualification analysis	9,411,178	3,950,120	722
Cross-race additive separability pairs	10,934,525	—	—
Boston-destination pairs	690,058	—	—

Notes: The raw linked panel is the web-scraped dataset after entity resolution. The AKM sample requires at least two appearances per regional runner, a known year of birth, and trims the slowest 0.1% within each race-year. The course speed index retains courses with at least 50 AKM observations after Empirical Bayes shrinkage. The qualification analysis uses all non-Boston U.S. road observations at indexed courses with BQ standards; its course count is smaller than the index because non-US and trail courses and courses without indexed years in the qualification window drop out. Additive-separability pairs link each runner's destination race to every prior marathon at a different course within 18 months; Boston-destination pairs restrict the destination to Boston. Samples are used as follows: the raw panel in Sections 4.4 and 5; the AKM sample in Sections 3 and 6; the qualification sample in Sections 4–6; the pair samples in Section 3.3 and the diagnostics of Section 3.4; and the geocoded US subpopulation in Section 4 (selection and choice analyses).

Different exercises use deliberately different "near-BQ" bandwidths, matched to each design:

Table C2: Near-threshold bandwidth conventions, by exercise

Exercise	Bandwidth	Rationale
Bunching density	± 5 min of posted standard	Spike width at the kink
Era sorting comparison	± 5 min of neutral margin	Symmetric around the target
Dissimilarity / excess sorting	$\pm 5\%$ of standard (neutral)	Proportional across demographics
Inframarginal control group	$> 15\%$ slower (neutral)	Course choice cannot affect Q_i
Choice-model pressure index	Tent peaked -8 min, width 10	Matches prior-best response peak

Because web-scraped results sometimes contain misaligned timing columns (chip times versus gun times), I also select the time variable for each race-year that minimizes within-runner prediction error across races.

C.1 Annual Summary Statistics

Table C3 reports annual panel statistics for every year in the sample. Columns show the total finisher count, the number of distinct courses, the share of rows with a known age or year of birth, the share female (conditional on a labeled gender), the count and population share of courses at least 2% faster than Boston, the count of apparent qualifiers, the share of finishers clearing their own age-gender BQ standard, and the posted and effective M18–34 BQ thresholds that governed Boston admission in the following year. The qualifying-share column reflects both the tightening BQ standard (M18-34 moves 3:10 $\rightarrow$ 3:05 $\rightarrow$ 3:00 $\rightarrow$ 2:55) and the slow drift in marathon-field composition.

C.2 Entity Resolution

I link runner records using a procedure that begins with a high-recall merge and then prunes implausible groupings using rarity and within-group contradictions. The first stage builds name-rarity tables (counts of observations sharing a given canonical first name, last name, gender, and year-of-birth bucket) so that downstream decisions can condition on how distinctive a given identifier combination is, and constructs a graph of marathons that share unusually many high-confidence linked runners; connected components of this graph define "communities" of races that recruit from overlapping runner populations. The second stage forms initial runner clusters by merging observations that share a canonical first name, last name, and gender, with year-of-birth resolved at the runner level: missing years of birth are imputed from age bands or from peer observations of the same canonical identity, and the merge requires that all observations sharing an identity have year-of-birth ranges that intersect. Country and state are imputed where partially observed, and runner-level country is propagated within the cluster.

The third stage splits clusters that look implausible. Hard-contradiction splitters separate observations within a cluster that have non-intersecting year-of-birth ranges, two finishes on the same date at different races, more than one finish at the same race in the same year, conflicting observed gender, or finishes labeled to incompatible countries. A rarity-based splitter then breaks up clusters whose name combinations are common enough that the high-recall merge is likely to have collapsed several distinct people, with rare combinations deferred to the next sub-stage. A connectedness-based splitter uses the marathon-community graph from the first stage: observations of a candidate identity that fall in disconnected race communities are split apart, since a single physical runner is unlikely to compete in entirely separate regional or competitive ecosystems. A final hard-validation sweep re-applies the contradiction tests on the post-split clusters. The output is a runner-id assignment for every observation in the panel. As a robustness check, I re-estimate the main course effects on the subsample of clusters that pass the strictest filters (rare name combinations, narrow year-of-birth windows, no high-volume aggregator flags). The resulting estimates have a Spearman correlation above 0.97 with the baseline estimates.

C.3 Time Variable Selection

Web-scraped race results sometimes contain misaligned columns, so a variable labeled "time" may correspond to gun time, chip time, or an unrelated field. For each race-year, I therefore select the time variable that minimizes within-runner mean squared prediction error, using the runner's times in other races as the prediction benchmark. Any residual mismatch between chip and gun time is small relative to the interquartile range of estimated course effects (11 minutes).

Appendix D Robustness

I estimate six alternative specifications. The first three alter the sample restrictions: excluding trail races, restricting to high-confidence entity-resolution matches, and requiring at least three appearances per runner. The fourth drops observations within ± 10 minutes of the BQ standard, isolating identification from runners far from the strategic-effort margin. The fifth replaces each finish time with its debunched value (subtracting the mass piled up at salient minute marks within each race-year). The sixth restricts to runners who placed top 5 in their age-gender group at any race—"race-to-win" finishers for whom BQ pursuit is not the operative incentive. Four of the six specifications yield Spearman rank correlations of 0.99 or higher with the baseline estimates (Table D1). The two that deviate most are the high-confidence entity-resolution restriction (Spearman 0.953, mean $|\Delta \text{FE}| = 2.3$ minutes), which discards most of the mover network, and the top-5 specification (0.969, 2.1 minutes), consistent with the elite tail of finishers being a different selection of runners than the BQ-pursuing population that dominates the baseline sample.

Robustness to Strategic-Effort Contamination. A concern for the behavioral estimates of Section 4—and for any use of the runner fixed effects as neutral-course baselines (Appendix F)—is that the runner fixed effects may themselves be contaminated by strategic effort near the BQ threshold. If runners systematically exert more effort when close to qualifying, their estimated baseline performance could be biased upward, which would in turn bias any residual built on it. I address this concern in two ways.

First, I re-estimate the AKM model after excluding runners within 5 or 10 minutes of the BQ cutoff. I also estimate the model on a sample restricted to inframarginal runners, defined as those whose predicted baseline performance is more than 5 minutes slower than the BQ threshold on a neutral course. In this sample, proximity to the threshold cannot affect effort. In all three cases, the estimated course effects are substantively unchanged relative to the baseline, indicating that the course-speed index is not driven by strategic effort.

Second, I re-estimate the full AKM specification after dropping all observations within ± 5 minutes of the BQ cutoff. This produces "clean" runner fixed effects identified only from observations for which runners had no immediate BQ incentive to exert differential effort. These fixed effects are highly correlated with the baseline estimates, confirming that runner-level neutral-course baselines are robust to strategic-effort contamination.

Debunched Finish Times. A further concern is that strategic bunching at BQ thresholds and round numbers mechanically inflates course fixed effects at races where bunching is concentrated. I address this by estimating a smooth counterfactual finish-time density outside an excluded bunching region around each threshold, identifying excess mass, and probabilistically reallocating bunching runners across the excluded region according to the counterfactual density. I run three asymmetric excluded windows as sensitivity: narrow $[-3, +5]$ minutes, medium $[-5, +7]$ minutes, and wide $[-7, +10]$ minutes. The debunched course effects are virtually identical to the baseline (Spearman ≥ 0.9997 for all three windows), indicating that bunching does not materially affect the estimated course speed index.

Non-Marginal Runners. To test whether course effects are driven by BQ-motivated effort, I identify runners who placed in the top K of their age-gender group at any race—runners who compete to win, not to qualify for Boston—and keep all their observations across all courses. The resulting course effects are highly correlated with the baseline (Spearman 0.969 for the top-5 variant; Table D1) and, if anything, show BQ-focused courses as faster, not slower, than the baseline estimates. This is inconsistent with BQ-related effort inflating the course fixed effects.

Course-Level Additivity Residuals. For each origin course r with at least 50 runner-pairs (i, r, Boston) where the same runner raced the origin course and Boston within 18 months, I compute the average log-time gap and compare it to the FE-implied gap. Let $g_{ir}^{\text{obs}} = \ln T_{i,\text{Boston}} - \ln T_{ir}$ denote the observed log-time difference (Boston minus origin) for runner i's pair and $g_{ir}^{\text{FE}} = \hat{\delta}_{\text{Boston}} - \hat{\delta}_r = -\hat{\delta}_r$ the corresponding FE-implied difference (since $\hat{\delta}_{\text{Boston}} = 0$ by normalization). If the AKM is perfectly additively separable, the expected value of g_{ir}^{obs} is g_{ir}^{FE} and the residual is mean-zero. The *course-level additivity residual* is the course-specific average of that residual, after partialling out year-to-year temperature deviations at the origin and at Boston:

$$\widehat{\Delta}_r \;=\; \mathbb{E}_{i\in\mathcal{P}_r}\Big[\big(\ln T_{i,\text{Boston}} - \ln T_{ir}\big) \;-\; \big(-\hat{\delta}_r\big)\Big] \;=\; \mathbb{E}_{i\in\mathcal{P}_r}\big[\ln T_{i,\text{Boston}} - \ln T_{ir}\big] \;+\; \hat{\delta}_r, \tag{8}$$

where $\mathcal{P}_r$ is the set of runners who raced both course r and Boston within the 18-month window. Reported in minutes at a 3:20 reference pace, $\widehat{\Delta}_r$ measures how much slower (or faster) runners actually performed at Boston than the origin course's fixed effect would have predicted. A perfectly additively separable index would have $\widehat{\Delta}_r = 0$ at every course; positive values indicate the FE overstated the origin course's speed (runners were slower at Boston than the FE implied), and negative values indicate the FE understated it.

Figure D2 plots $\widehat{\Delta}_r$ against the origin-course FE for 345 courses with $|\hat{\delta}_r| \leq 25$ minutes. Most courses sit inside a ± 5 minute reference band around zero, confirming that the FE index is additively separable to within a few minutes course by course. Most of the small positive shift in the mean (+2.5 minutes) reflects a Boston-specific slowdown common to all origin courses—travel fatigue, the April race date, and the Boston course's own well-known difficulty—that the FE normalization to Boston cannot absorb at the individual-runner level. The flat BQ.2 courses show near-zero gaps, confirming that their fast fixed effects accurately predict Boston performance.

Table D1 summarizes Spearman rank correlations across all specifications. Figure D1 plots all specifications for key races.

Table D1: Spearman rank correlations with baseline course fixed effects across robustness specifications. Mean $|\Delta\text{FE}|$ reports the average absolute change in estimated course speed (minutes at 3:20 pace) relative to the baseline.

Specification	N Races	Spearman ρ	Mean $\lvert\Delta\text{FE}\rvert$ (min)
Sample restrictions			
Exclude trail races	731	0.999	0.2
High-confidence ER only	721	0.953	2.3
Min 3 appearances	799	0.998	0.4
Strategic-effort robustness			
Excl. BQ margin ±10 min	804	0.994	0.8
Debunched finish times			
Debunched (medium $[-5, +7]$)	815	1.000	0.1
Non-marginal runners			
Top 5 by age group	804	0.969	2.1

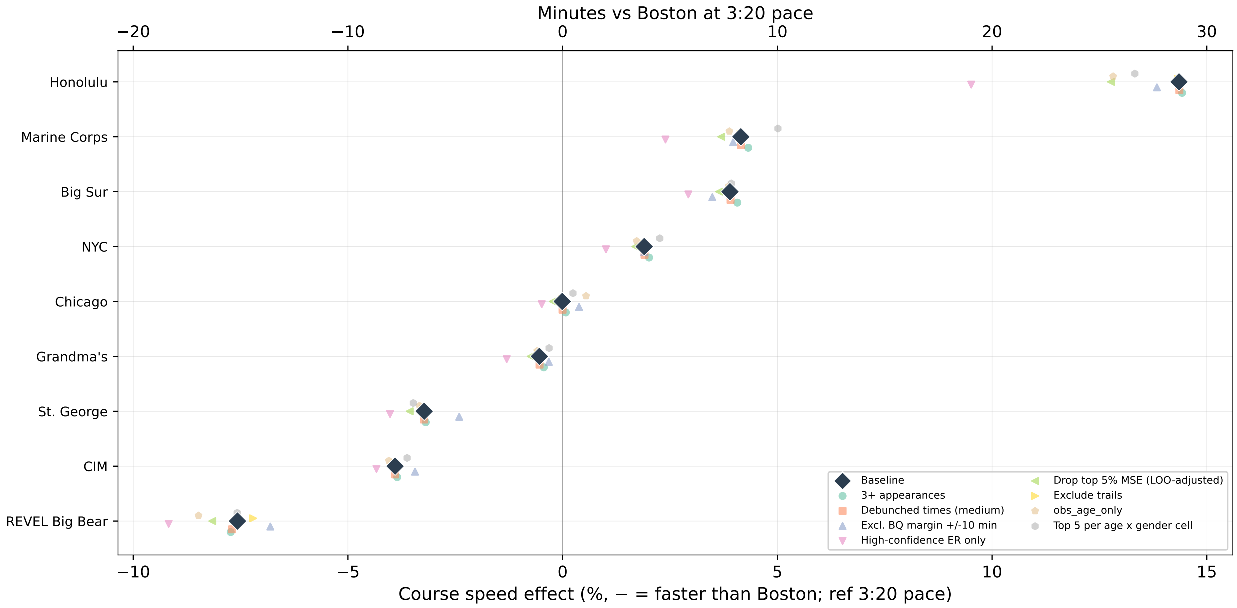


Figure D1: Course speed effects for key races across robustness specifications. Diamond: baseline AKM estimate. Small markers: alternative specifications including sample restrictions, BQ-margin exclusions, debunched finish times, and non-marginal runners.

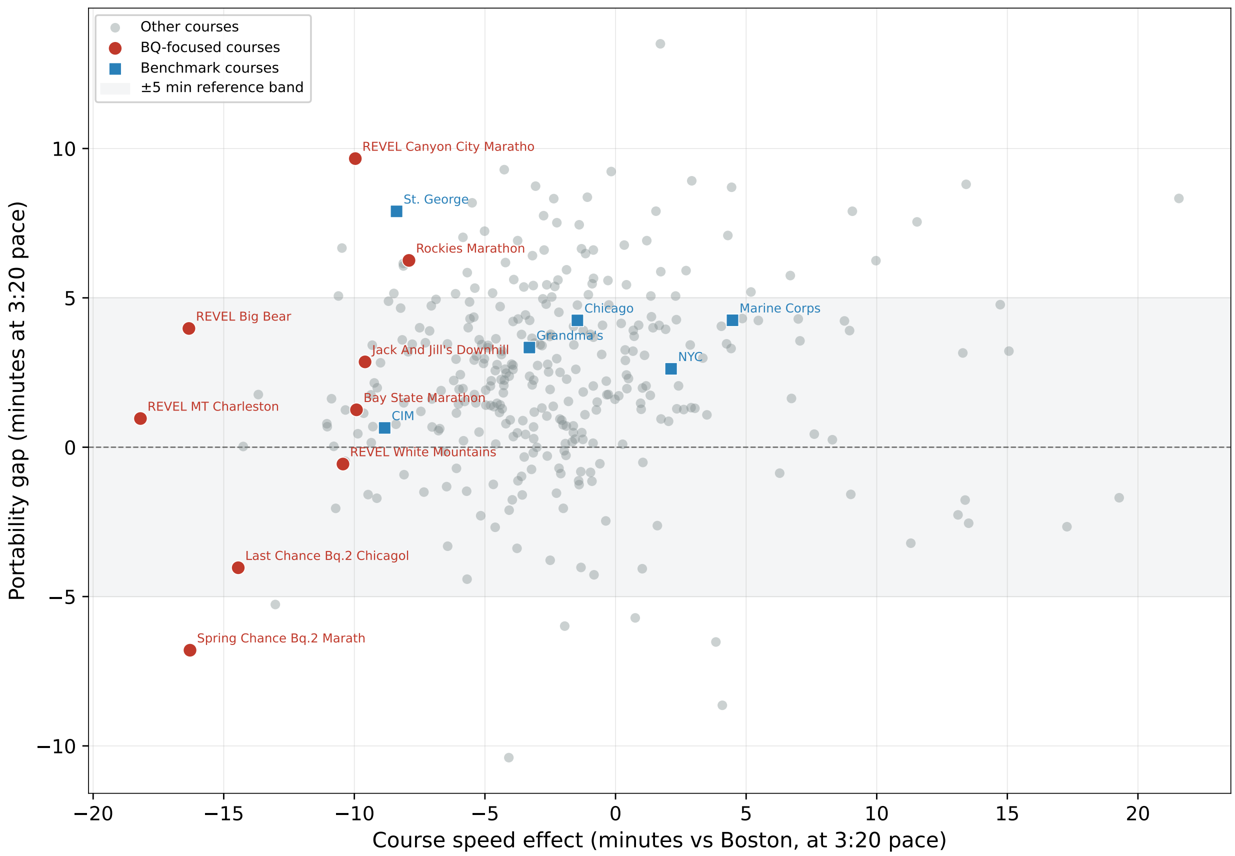


Figure D2: Course-level additivity residual $\widehat{\Delta}_r$ (equation (8)) plotted against the origin-course FE. Each point is a qualifying course with at least 50 same-runner pairs who subsequently raced Boston within 18 months (weather-adjusted, restricted to $|\hat{\delta}_r| \leq 25$ minutes). Perfect additive separability implies $\widehat{\Delta}_r = 0$; positive values mean the FE overstated the origin course's speed (runners slowed more at Boston than expected), negative the reverse. Most courses sit inside the ±5 minute reference band. Labeled outliers: Brooklyn Marathon (−18 min) has known course-measurement history (alternative paths through Prospect Park in different years), so Brooklyn times look slower than they were and Boston appears artificially fast by comparison. Mt Hood (+11 min) and Clearwater (+10 min) are small, goal-race courses whose co-racing subsample is heavily peaked—runners hit peak performance at the goal race and regress at Boston. LA Marathon (+8 min) has the highest precision in the figure (N = 6,128) and its FE appears to underweight the Boston-specific heat slowdown because LA itself is a mild-weather race. Spring Chance BQ.2 sits just below zero, consistent with its BQ-focused design producing a small additional peaking effect at the origin that does not fully propagate to the Boston run.

Robustness of the Misallocation Rate Series. The performance-level headline counts race-results, so a runner with multiple qualifying races can contribute more than once. Figure D3 therefore collapses to one runner–cohort observation. A unique runner is course-dependent only if the runner has an observed posted-standard qualifier and no observed Boston-equivalent posted-standard qualifier anywhere in the window; the earlier "any course-dependent race" rule is not used. The broader sample treats unindexed-course qualifiers as unclassified rather than using them to establish participant course dependence. The entrant-level effective- cutoff series in the main figure is stricter still and is the statistic used for claims about the realized Boston field.

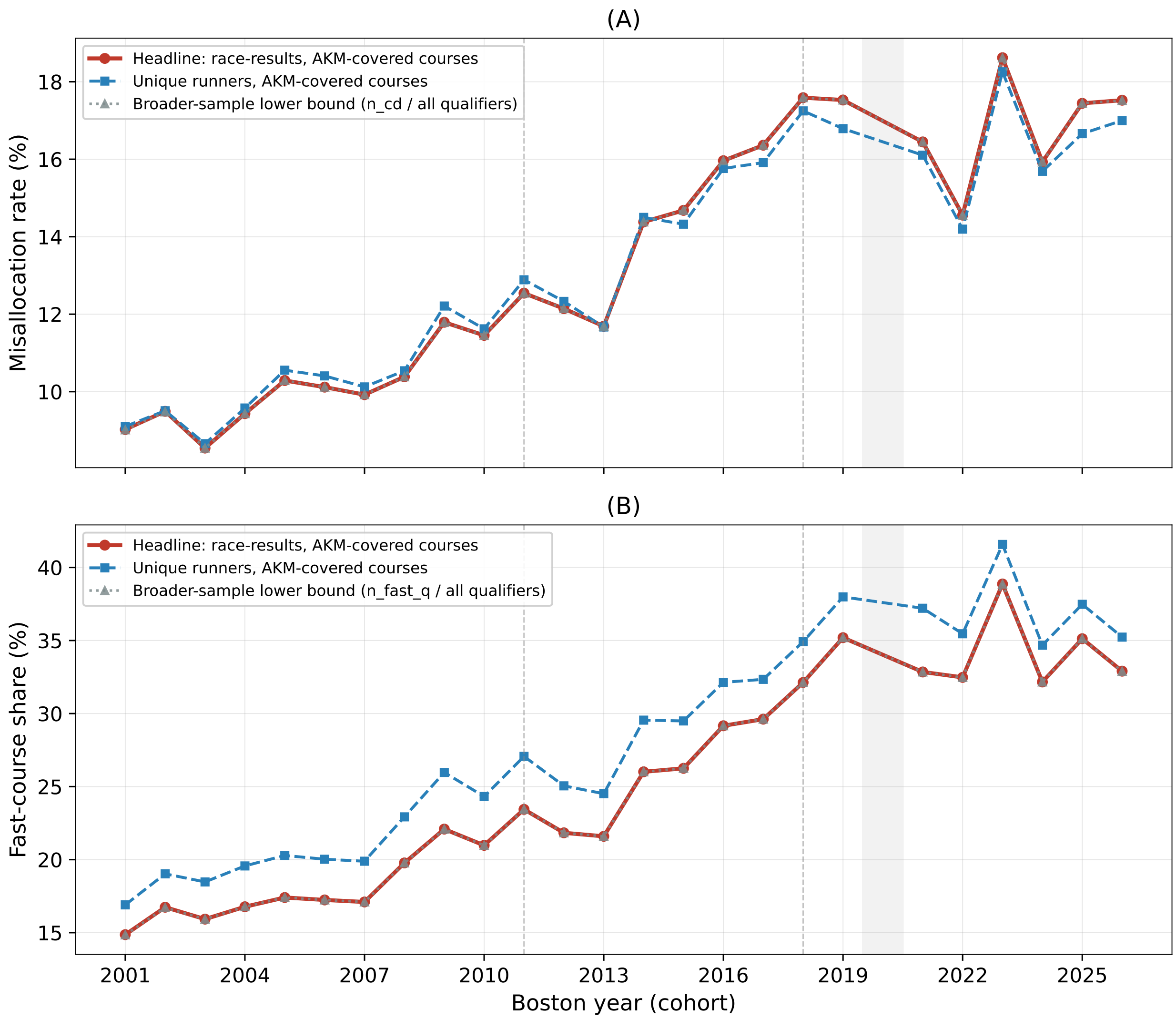


Figure D3: Robustness of the performance-level misallocation series. The unique-runner series counts a runner as course-dependent only when none of the runner's observed qualifying-window performances is a Boston-equivalent posted-standard qualifier. The broader-sample line expands the denominator to qualifiers at unindexed courses, which remain unclassified. Boston 2000 is excluded; Boston 2020 is shaded; dashed lines mark standard changes.

Exhibits Supporting the Main-Text Series. The following exhibits supplement Sections 5, 4, 4.4, and 6: the qualifying-menu mechanics, the near-BQ versus inframarginal sorting series, the estimand-reconciliation and predictable-pressure tables, the excess-mass detail, the full decomposition chains by cohort with the alternative-baselines grid, the course-level over-attraction ranking, and the displacement comparison across every index and rule.

Table D2: Reconciling the sorting magnitudes: each estimate, its estimand, its population, and the channels it includes.

Estimate	Value	Estimand	Population	Channels
Excess fast-course share, near-BQ	+7.9 pp	Stock: fast-course share of near-BQ finishers minus the era's full field	Near-BQ finishers (neutral ±5 min), 2013–2019	All BQ-related sorting (venue choice + correlated effort)
Announcement difference-in-differences	+3.2 pp (SE 0.2)	Change: differential fast-course choice of near-BQ runners after the 2011 tightening, net of year and runner controls	US runners with geocoded home	Venue choice (demand) response to the announcement
Choice-model counterfactual	+11.7 per 1,000	Flow in qualification units: marginal qualifications produced by the pressure-induced choice shift, course menu fixed (partial equilibrium)	Near-BQ runner-occasions in the choice sample	Pressure $\times$ speed channel of the conditional logit only
Misallocation rate (screen side)	10% $\rightarrow$ 18%	Stock: share of apparent qualifiers whose qualification depends on the course they ran (2004 $\rightarrow$ 2019 cohorts)	Apparent qualifiers at indexed courses	Outcome of all channels combined, screen side

Table D3: Selection on predictable pre-race state: pressure, venue speed, and the announcement difference-in-differences.

	(1)	(2)	(3)	(4)	(5)
Dep. variable:	Course speed	BQ pressure	Fast course	Qualifies	Boston take-up
	(log FE)	(index 0–1)	(pp)	(pp)	(pp)
BQ pressure	-0.0151		11.19	24.27	29.37
Chosen-course speed (log FE)		-0.5801			
Near BQ ($\pm$5 min)	0.0059		-5.47	-9.22	-6.35
Near BQ $\times$ post-2011	-0.0024		3.18	0.20	-3.79
Post-2011	-0.0046	-0.0051	5.81	-1.21	-1.59
Log distance to course	-0.0002	0.0039	0.71	0.35	0.48

Table D4: Polynomial excess-mass estimates at the BQ threshold and round-number placebos, by course type and era.

	Pre-2012	2012+
Excess mass at the BQ threshold		
All courses	2.14	1.36
Fast courses	3.15	1.85
Medium courses	2.08	1.18
Slow courses	1.49	1.06
Excess mass at the 4:00 round-number placebo		
All courses	1.83	1.96
Fast courses	2.05	1.71
Medium courses	1.86	2.07
Slow courses	1.66	1.80

Table D5: Channel contributions to the 2019 misallocation rate (pp) under alternative assignment baselines and path orderings. Sorting is invariant to the ordering; reordering reallocates only between the threshold and supply channels. Residence-subsample rows are comparable to each other, not to the full-sample rows.

Baseline	Ordering	Threshold	Supply	Sorting
Random (field shares)	Threshold first	2.3	5.1	5.5
Random (field shares)	Menu first	3.7	3.7	5.5
Inframarginal (|margin| $>$ 20 min)	Threshold first	2.1	4.5	6.6
Inframarginal (|margin| $>$ 20 min)	Menu first	3.4	3.3	6.6
Inframarginal (|margin| $>$ 15 min)	Threshold first	2.1	4.7	6.4
Inframarginal (|margin| $>$ 30 min)	Threshold first	2.1	4.4	7.0
Inframarginal, gender-conditioned	Threshold first	2.1	4.6	6.6
Random, residence subsample	Threshold first	2.6	5.1	7.2
Gravity (distance decay), residence subsample	Threshold first	2.6	4.8	7.8
Gravity (distance decay), residence subsample	Menu first	4.1	3.3	7.8

Table D6: Displacement under alternative indices and rules (fixed field, US 2018–19).

Policy	Displaced (%)	From fast courses (%)	Runners out	Runners in
Common threshold (status quo)	0.0	–	0	0
Behavior-adjusted (headline)	13.0	89.2	11,045	11,045
AKM physical (naive ψ, starting point)	13.7	89.6	11,644	11,644
Clean AKM (excluded-margin)	13.4	89.4	11,366	11,366
Routine-attempt (visitor control)	11.2	85.6	9,476	9,476
Reverse-pair (symmetric)	14.0	86.8	11,900	11,900
Observable-based (terrain)	9.7	63.4	8,234	8,234
Conservative shrinkage ($k = 1.645$)	11.1	92.7	9,411	9,411
BAA 2027 net-drop penalty	2.1	–	1,746	1,746
Symmetric elevation formula (rule)	7.5	–	6,367	6,367

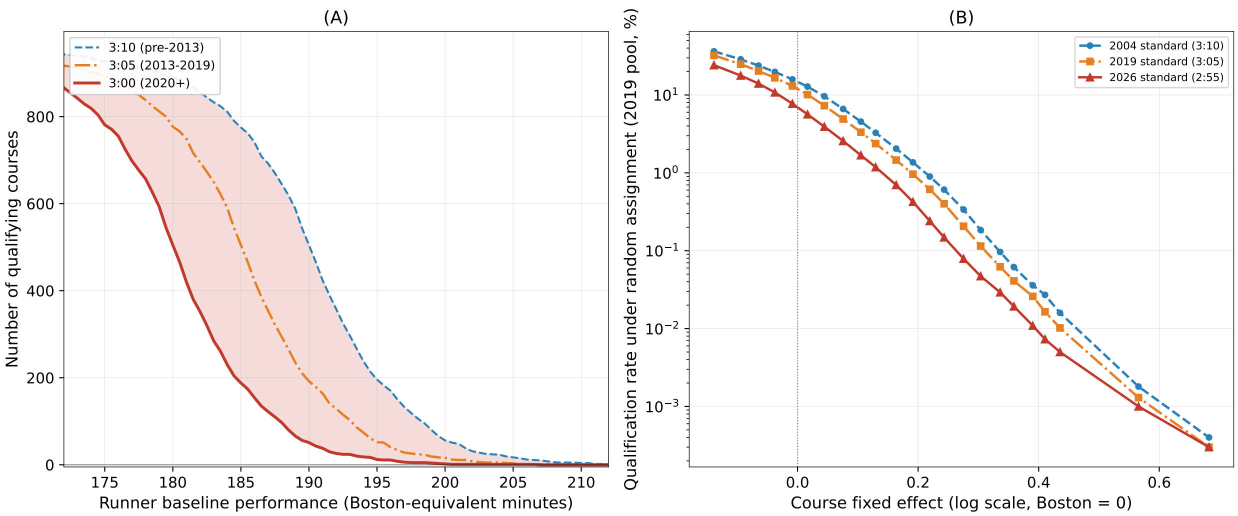


Figure D4: Qualifying mechanics. Panel A: the number of indexed marathons at which a runner of given neutral ability can qualify, under three historical standards. Panel B: qualification rates by course speed under the same three standards.

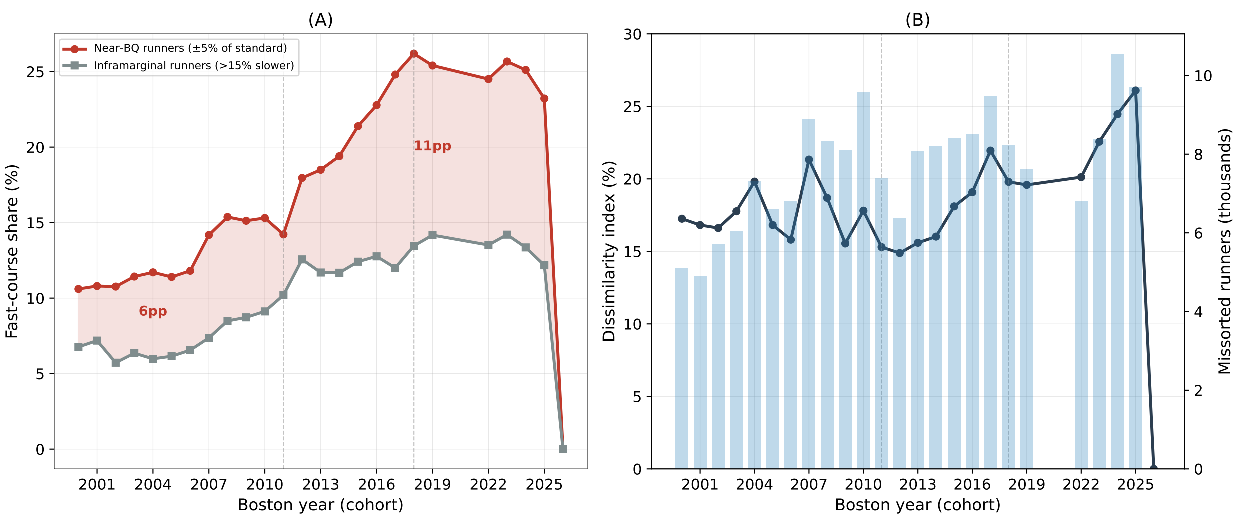


Figure D5: Near-BQ vs. inframarginal fast-course shares (Panel A) and the implied excess sorting (Panel B), 2000–2025. Under the assumption that inframarginal runners supply the no-incentive counterfactual distribution, roughly one in five near-BQ runners is at a course they would not otherwise have chosen.

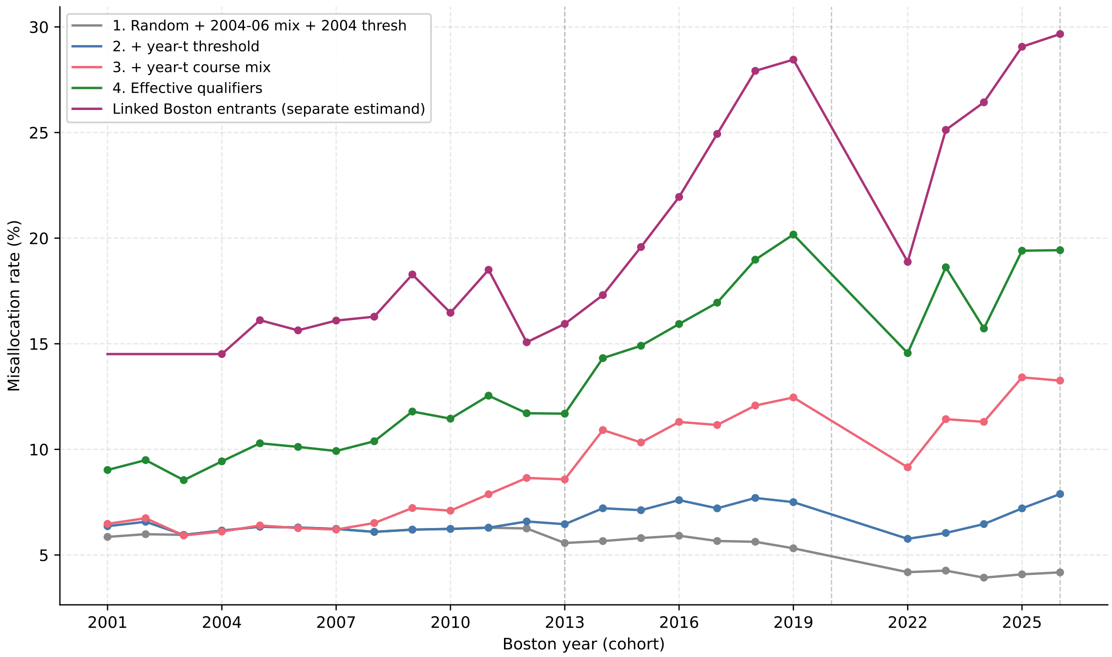


Figure D6: Counterfactual accounting of the misallocation rate, Boston 2001–2026 cohorts. Steps 1–4 add the effective cutoff, course menu, and actual assignment to the random baseline. The separate linked-entrant line reports the direct Boston-field measure and is not an additive take-up channel. Vertical lines mark the first cohort under each new standard.

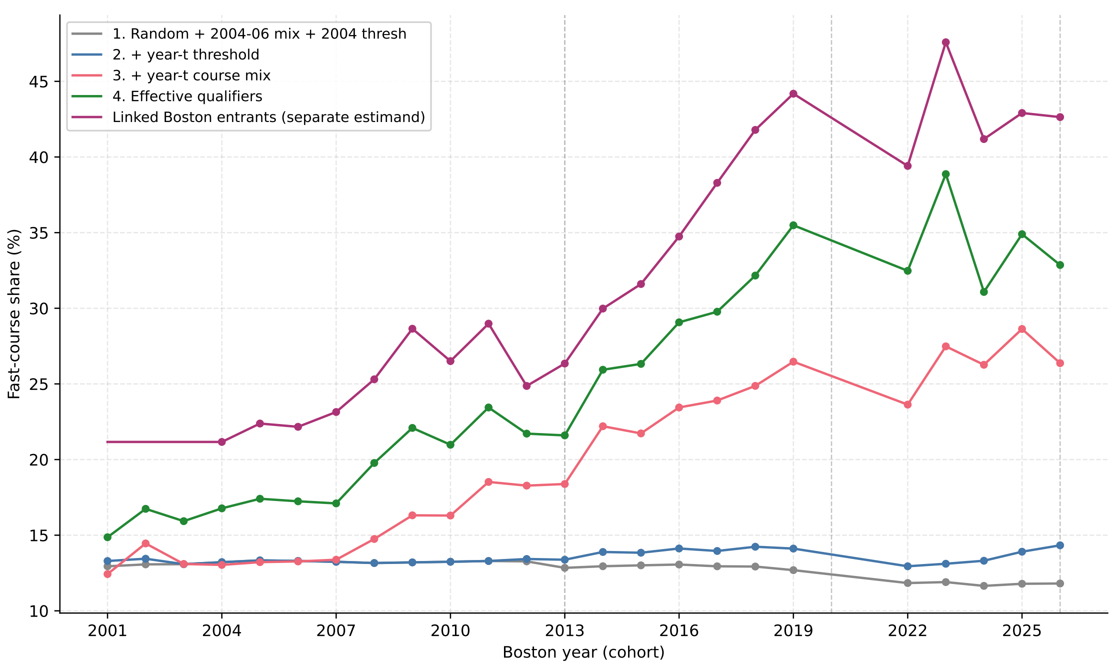


Figure D7: The same decomposition for the fast-course share among apparent effective qualifiers (steps 1–4) and the separate linked-Boston-entrant series.

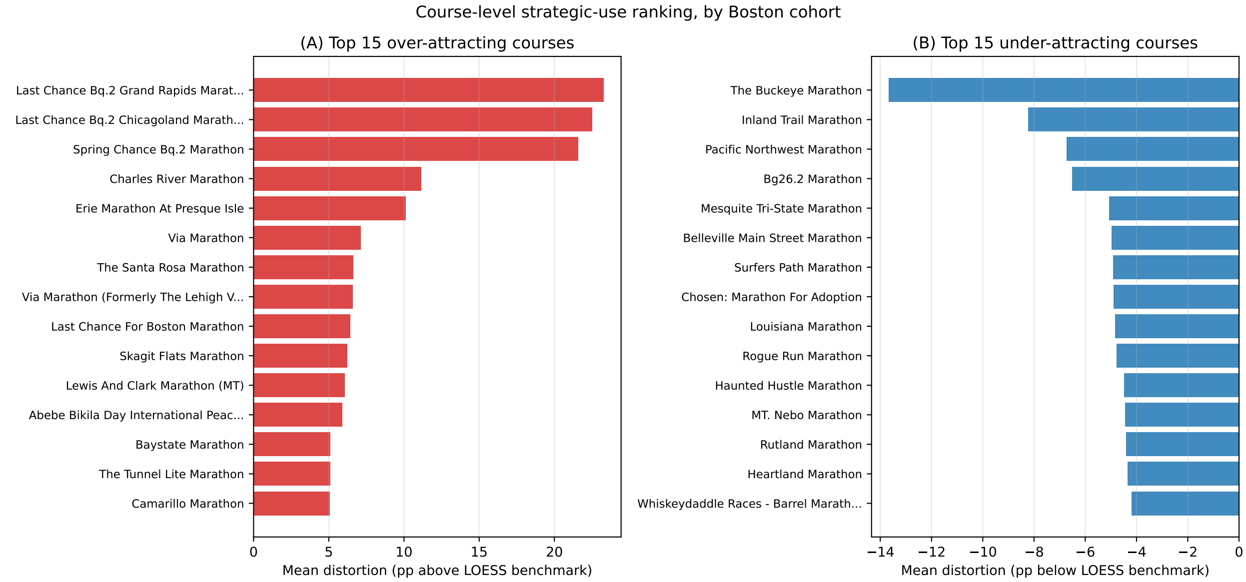


Figure D8: Top over- and under-attracting courses, ranked by cohort-weighted deviation from the smooth speed benchmark. Dedicated BQ products lead the over-attractors.

The Full Corrected-Index Suite. Table D7 reports the complete seven-index suite behind the four-index main-text version (Table 1): the three variants demoted from the main text are the clean re-estimate (excluding near-threshold observations), the reverse-pair symmetric estimator, and the conservative shrinkage rule. The displacement band quoted throughout (9.7–14.0%) spans all seven.

Table D7: The full course-speed index suite: construction, what each neutralizes, and the implied fixed-field displacement (US 2018–19 qualifiers, field size 84,925).

Index	Neutralizes	Displaced (%)
Behavior-adjusted (headline), eq. (4)	Course-level peaking of strategic users	13.0
Naive AKM ψ (starting point)	—	13.7
Clean AKM (excluded margin)	Strategic effort near the threshold	13.4
Routine-attempt	Visitor/goal-race composition	11.2
Reverse-pair (symmetric)	Direction-specific order effects	14.0
Observable-based (terrain)	All non-physical variation (and some physical)	9.7
Conservative shrinkage (k=1.645)	Estimation noise (zeroes imprecise courses)	11.1

Appendix E AKM Diagnostics

Following Card, Heining and Kline (2013), I assess the AKM decomposition on two dimensions: year-to-year stability of course effects within each race, and nonseparability of runner and course effects. Additive separability (Section B.4) already provides the mover-style cross-venue validation that is typically the third CHK diagnostic. These exercises are intended as stress tests of the additive approximation rather than as alternative estimators of the main course index.

Within-Course Year-to-Year Stability. If the estimated course effect captures a persistent feature of the race, the year-to-year variation in finish times at a given course, net of common year trends and runner composition, should be small relative to the cross-course variation in the index. I measure this stability using the standard deviation of race-year mean residuals within each course. After the AKM, for each race-year cell I compute $\bar{e}_{r,t} = \text{mean}(\hat{\varepsilon}_{i,r,t})$; the within-course across-year SD of $\bar{e}_{r,t}$ captures year-to-year course variation net of the common year FE.

Across the 691 courses with at least three years of data, the median within-course across-year SD is 2.8 minutes (at a 3:20 pace), the 95th percentile is 6.7 minutes, and the mean is 3.3 minutes. These are small compared with the cross-course SD of the permanent course effect itself (on the order of 20–30 minutes), implying that $\hat{\delta}_r$ is well identified as a time-stable course attribute. Figure E1 plots this within-course across-year SD against the permanent course effect for each race, highlighting benchmark courses. Marine Corps (1.9 min), St. George (2.2 min), and Honolulu (2.2 min) are among the most stable despite spanning a wide range of course speeds; Boston itself is at 3.7 minutes, reflecting genuine year-to-year variation in weather and field composition.

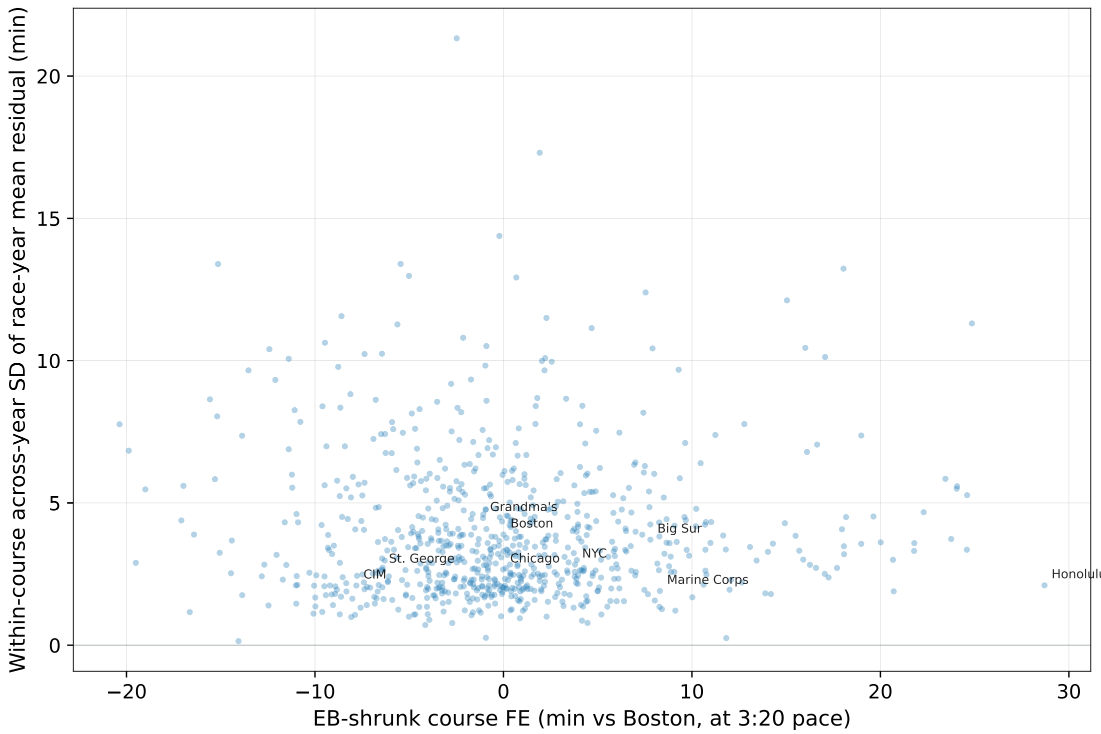


Figure E1: Within-course year-to-year stability of the AKM course effect. Horizontal axis: EB-shrunk course FE $\hat{\delta}_r^{shr} = w_r \hat{\delta}_r$ in minutes vs Boston at 3:20 pace ($|\hat{\delta}_r^{shr}| \leq 25$ min plus benchmarks always retained). Vertical axis: standard deviation of race-year mean residuals within that course, net of the common year FE, the runner and course FEs, the age-gender polynomial, and year-to-year deviations in morning temperature. Humidity, wind, and precipitation are not in the AKM and therefore contribute to this SD. The median across-year SD is 2.8 minutes and the 95th percentile is 6.7 minutes, small relative to the cross-course FE range. Empirical Bayes shrinkage has already pulled the volatile courses toward zero: the weights range from 0.73 at the noisiest course (Waterloo, which has only 3 years and 59 observations) to 0.999 at Boston, Chicago, and NYC; the largest shrinkages are between 10 and 27 percent. The remaining labeled high-SD outliers have thin samples (Waterloo, Groundhog, Loonies Midnight, and Bay of Fundy International each have fewer than ten race-years and under 400 total finishers, so the race-year mean residual is noisy even after shrinkage) or weather-sensitive designs (REVEL Rockies is a high-altitude downhill race with documented year-to-year variation in wind and precipitation beyond what temperature alone captures). None of these represent identification failures of the AKM; they are cases where either the cell sample is thin or the non-temperature weather draw is genuinely volatile.

Nonseparability. Under the additive AKM structure, the course effect shifts all runners' log times by the same amount, so mean residuals should be approximately zero in every runner-ability × course-speed cell. To test this, I assign each observation to a runner-FE decile and a course-FE decile and compute the mean residual in each of the 100 cells. I then two-way demean to isolate the runner × course interaction: $\hat{\eta}_{rc} = \bar{e}_{rc} - \bar{e}_{r\cdot} - \bar{e}_{\cdot c} + \bar{e}_{\cdot\cdot}$. Under the additive model, $\hat{\eta}_{rc} \approx 0$ for all cells.

Figure E2 displays the interaction heatmap. The largest cell deviation is 0.61 log points × 100 (approximately 0.6% of finish time), and the mean absolute deviation is 0.10%. The interaction terms do exhibit some systematic structure, but it is concentrated in extreme runner-course cells and modest relative to the main course effects. In particular, the largest deviations occur in the tails of the runner and course distributions rather than in the middle of the support where the BQ-margin analysis is concentrated. These results indicate that the additive structure provides a useful approximation for road marathon performance, while leaving room for modest tail heterogeneity.

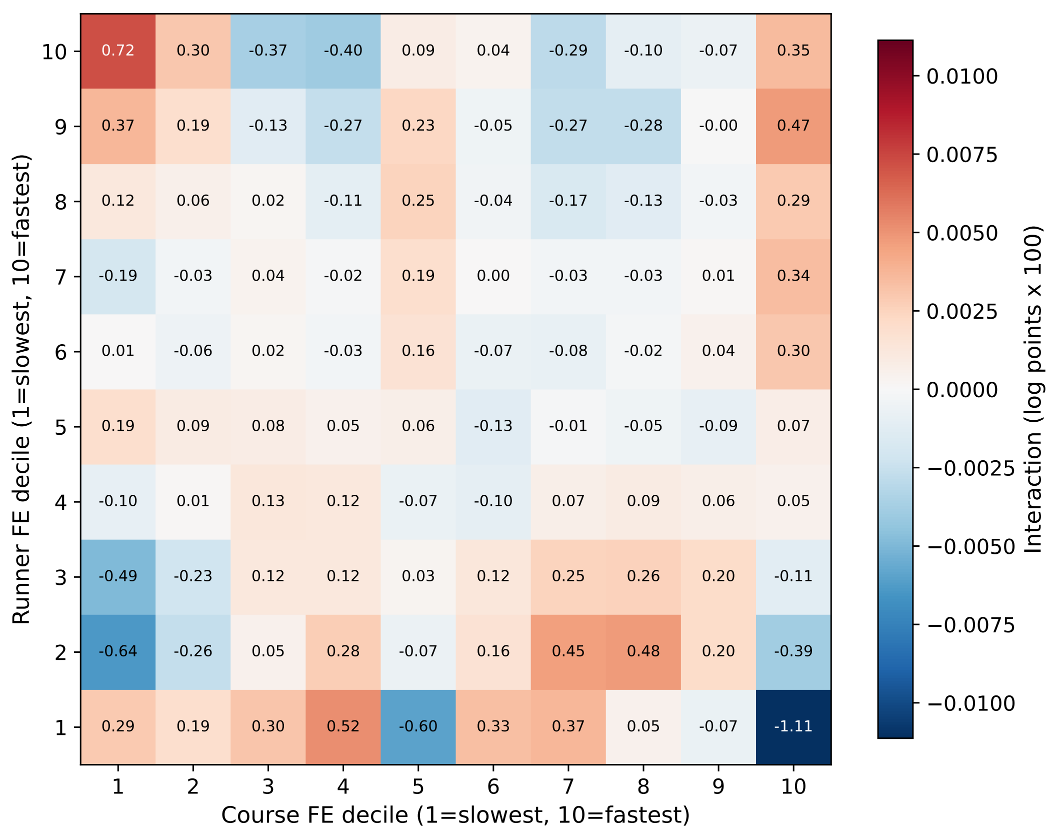


Figure E2: Nonseparability test: two-way demeaned interaction residuals by runner-FE decile (rows) and course-FE decile (columns). Values are log points × 100 (approximately percent). Under the additive AKM model, all cells should be near zero.

Appendix F Additional Bunching Diagnostics

This appendix collects supporting exercises for the demand-side evidence in Section 4: two bunching checks, a cross-margin pressure-index calibration, and the age-group notch analysis. The main-text figure 4 establishes that bunching is BQ-specific, that the spike tracks the qualifying standard as it moves, and that excess bunching on fast courses has grown over time. Here I report two further checks: whether the bunching could reflect the underlying ability distribution rather than behavior, and a granular version of the generic-versus-BQ comparison.

The AKM runner fixed effects, converted into predicted finish times on a neutral course, imply a baseline performance distribution that is smooth through salient round-number thresholds (Figure F1, Panel A). Actual finish times, by contrast, display pronounced spikes at half-hour and hour marks (Panel B). This contrast indicates that the bunching documented in Section 4 reflects behavior rather than irregularities in the underlying ability distribution.

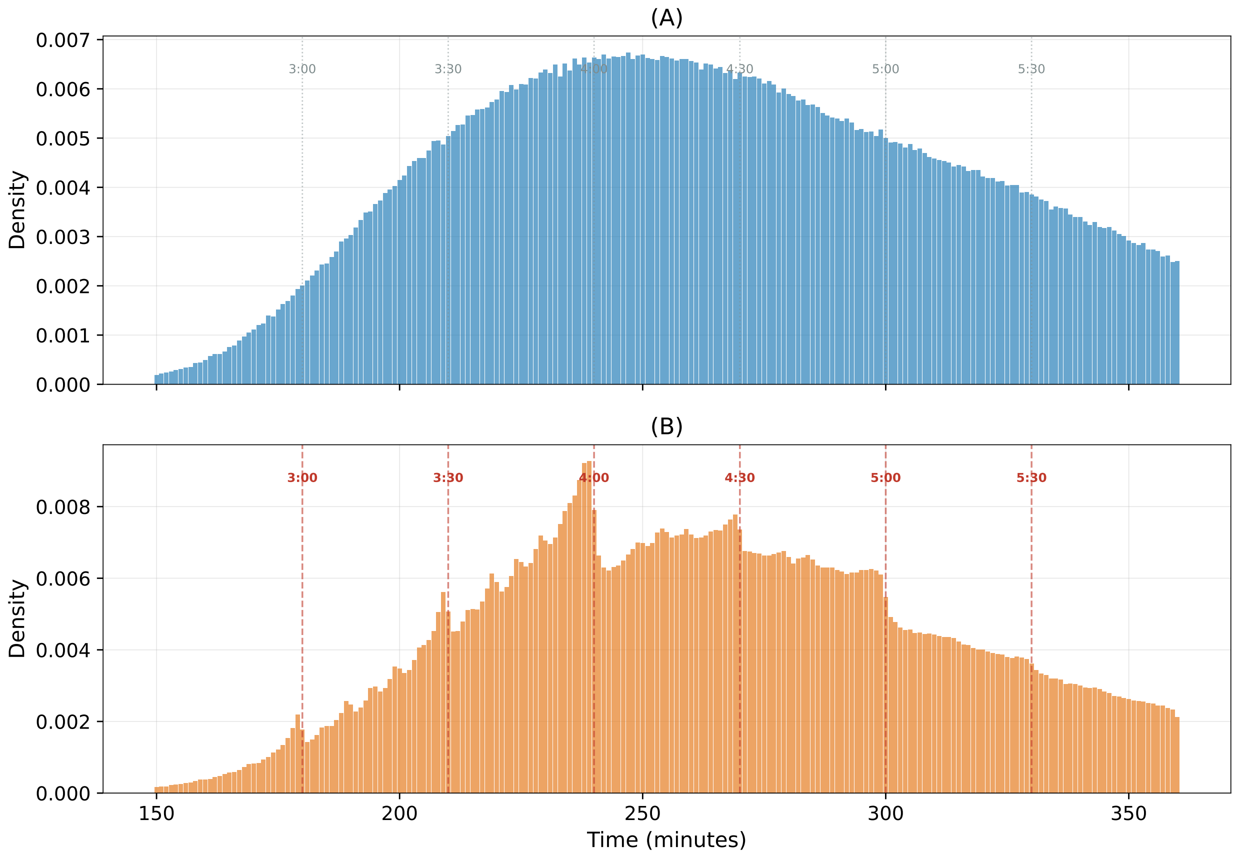


Figure F1: Baseline runner performance and actual finish times. Panel A plots the distribution of predicted neutral-course finish times. Panel B plots the distribution of actual finish times. The baseline distribution is smooth, whereas actual finish times bunch at salient thresholds.

Figure F2 provides additional detail on the contrast between generic round-number bunching and BQ-specific sorting. The top row examines fast-course share and finish-time density around the 4:00 mark (excluding runners for whom 240 minutes is a BQ threshold). The fast-course share is essentially flat across 4:00 (25.0% just above the mark, 26.4% just below), whereas it rises sharply at the BQ threshold (34.7% just above, 38.6% just below). The moving-BQ exercise in the main text (Figure 4) summarizes the same pattern.

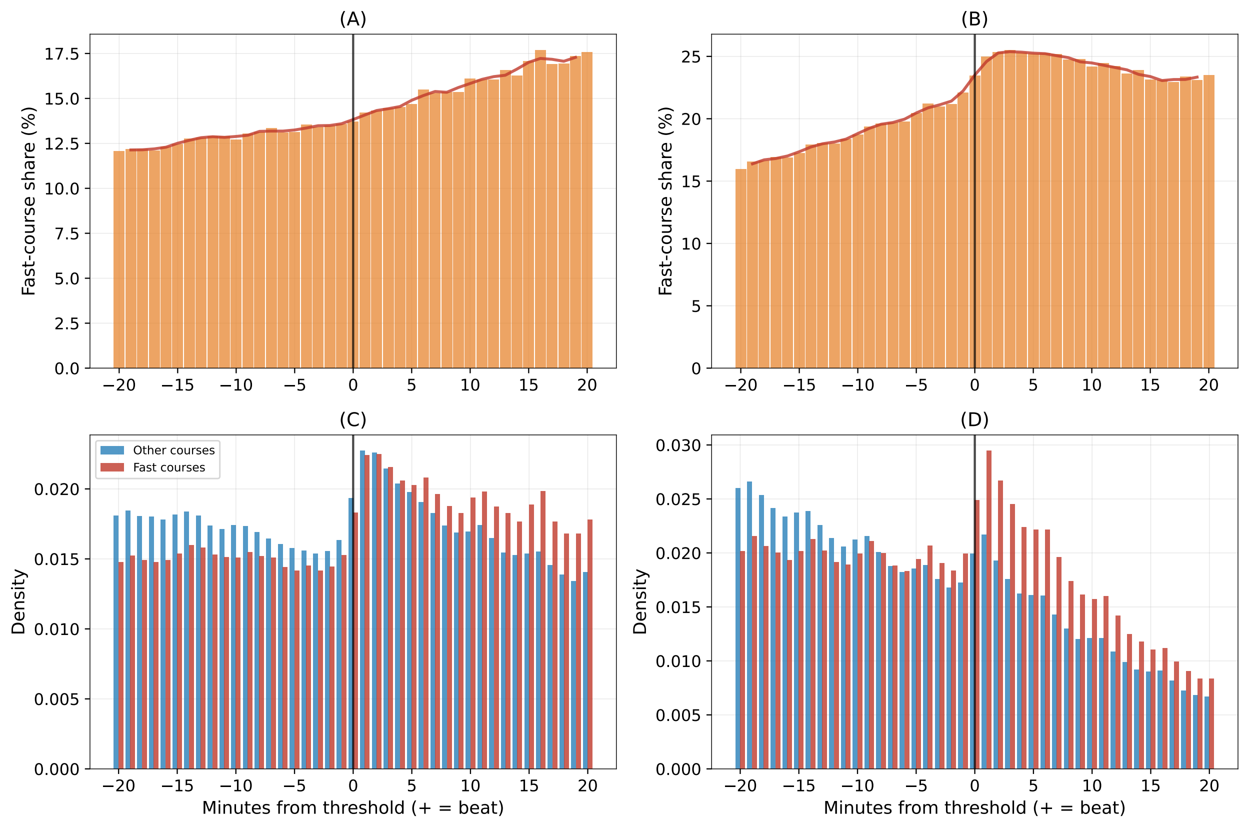


Figure F2: Generic round-number bunching and BQ-specific sorting. The top row examines the 4:00 mark, excluding runners for whom 240 minutes is the BQ threshold. The bottom row examines the BQ threshold. Sorting into fast courses is limited at the generic round number but pronounced at the BQ cutoff.

A Common Pressure Index Across Behavioral Margins. A constructed pre-race BQ-pressure index orders the sorting and bunching moments in the predicted direction in a cross-channel leave-one-out calibration, while the peaking-by-origin cross-section is dominated by course composition (fast, competitive origins draw consistent runners who fade less), even though the individual qualifier-peaking effect is solidly identified in Table B5. Because one of the three moments does not aggregate cleanly, the main text cites this calibrated consistency check only as corroboration; the underlying estimates are in the replication archive.[5]

[5]Full simulated-likelihood estimation of the joint model, with standard errors on the single-index restriction, is a natural next step and is beyond this paper.

F.1 The Age-Group Notch: Two Margins, Both Bounded

Qualifying standards relax discontinuously at five-year age boundaries, and the applicable category is determined by age *on Boston race day*— predetermined relative to any qualifying attempt—so threshold pressure is relieved exactly at aging-up. Two margins could respond, and the data bound both. The venue-choice margin cannot be estimated here: the data record year of birth but not birth date, and fall qualifiers map to a Boston nearly two years later, so a runner's Boston-day age is identified only to within about two years, and the boundary-adjacent cells—precisely where the response would appear—are the cells that cannot be classified with certainty. The attempt-timing margin can be tested despite the fuzziness: a runner who ages up between Boston year B and $B{+}1$ can delay a qualifying attempt past the mid-September window boundary so that it counts under the easier standard, and that retiming is detectable at the year level. The test returns a precise null: among near-margin runners who surely age up, the shift in post-boundary attempt share is 0.8 percentage points, with a confidence band of roughly ± 1 points after accounting for classification dilution, and a far-from-margin placebo attributes the remaining small contrasts to age-trend artifacts rather than incentives. The notch design therefore neither supports nor refutes the venue-choice prediction, the retiming channel is empirically absent, and no exhibit in the paper rests on either margin.

Table C3: Annual Summary Statistics – US Marathon Panel

Year	Finishers	Courses	Age%	F%	Fast courses	Fast%	App. Q.	BQ%	BQ_{M18} (min)	$Eff._{M18}$ (min)
2000	295,449	210	87	37	30	7.1	29,043	9.8	190	190.0
2001	272,053	217	92	36	28	7.7	24,597	9.0	190	190.0
2002	314,800	183	94	39	24	6.5	33,039	10.5	190	190.0
2003	326,159	205	93	39	28	7.0	31,423	9.6	190	190.0
2004	357,300	243	99	40	35	7.1	29,507	8.3	190	190.0
2005	380,656	262	98	40	36	7.0	33,924	8.9	190	190.0
2006	396,062	285	99	40	36	7.3	41,338	10.4	190	190.0
2007	401,342	296	99	39	41	8.5	41,355	10.3	190	190.0
2008	414,144	312	95	41	52	9.2	44,320	10.7	190	190.0
2009	460,915	313	95	40	50	9.5	53,471	11.6	190	190.0
2010	493,324	304	100	41	51	10.0	52,786	10.7	190	190.0
2011	491,368	337	98	41	59	10.4	54,228	11.0	190	188.8
2012	473,579	367	90	42	67	12.2	35,550	7.5	185	185.0
2013	529,993	379	94	43	70	12.1	47,967	9.1	185	183.4
2014	544,236	386	98	44	73	12.1	49,016	9.0	185	184.0
2015	505,689	366	97	44	77	13.1	42,583	8.4	185	182.5
2016	496,042	366	94	44	78	13.3	38,378	7.7	185	182.8
2017	469,389	341	99	44	77	13.5	37,430	8.0	185	181.6
2018	448,391	346	93	44	82	14.6	42,922	9.6	185	180.1
2019	438,014	330	93	42	78	14.7	35,747	8.2	180	178.3
2020	40,652	41	98	43	13	11.6	2,138	5.3	180	172.2
2021	208,172	218	96	41	52	17.4	23,788	11.4	180	180.0
2022	354,366	277	96	41	62	14.5	40,571	11.4	180	180.0
2023	406,683	300	95	40	69	15.0	43,228	10.6	180	174.5
2024	478,421	405	99	40	78	14.5	45,219	9.5	180	173.2
2025	468,708	329	90	40	65	12.8	47,398	10.1	175	170.4

Panel covers race years 2000–2025; 2020 reflects near-total race cancellation due to COVID-19. Fast courses defined as $\widehat{\delta}_r < \log(0.98) \approx -0.0202$ (approximately 2% faster than Boston). Apparent qualifiers are finishers whose actual time cleared their age-gender cell's nominal BQ standard in force for the following April's Boston Marathon. BQ_{M18} is the nominal M18-34 standard; $Eff._{M18}$ is the effective (field-size-adjusted) M18-34 cutoff for the following April's Boston field. The most recent race-year row therefore pairs with the next Boston cohort; the 2025 row pairs with Boston 2026, whose admission cutoff was announced on September 23, 2025 as 2:50:26 (BQ 2:55:00 minus a 4:34 admission buffer). The BAA reported 33,249 qualifier applications, 24,362 accepted, and 8,887 rejected for the 130th Boston (Boston Athletic Association, 2025*b*). 2003 coverage reflects a scraping gap being filled in ongoing data-collection work.

Appendix G The Course-Choice Model

A conditional-logit model estimates venue selection directly. Each choice occasion is one non-Boston US marathon finish by a US-residence runner with at least one prior observed race; the choice set is the course actually run plus nine other US courses drawn uniformly at random from those active in the same calendar year (under uniform sampling of alternatives the conditional logit remains consistent), estimated on a random subsample of 299,938 occasions contributed by 221,682 distinct runners. Utility falls in a fixed long-distance travel cost and rises in the interaction of pre-race BQ pressure with course speed; the key coefficient is $\hat{\rho} = -6.10$ on pressure $\times$ speed, robust to entering distance as a long-trip dummy, continuously, or not at all (Table G1; standard errors clustered by runner). The policy-relevant translation is not the average shift in chosen-course speed, which is mechanically small, but the change in *qualification probability* the pressure-induced choice produces: switching off the pressure channel and re-solving the model, the response is concentrated among runners within a few minutes of the threshold and vanishes in the tails; the pressure channel accounts for about 11.7 marginal qualifications per 1,000 near-BQ runners in partial equilibrium with the course menu held fixed. One measurement note: the pressure variable is built from prior-best margins neutralized with $\hat{\delta}$, so estimation error enters both sides; results are robust to using raw prior times relative to the standard.

Table G1: Course-choice conditional logit: venue-selection response and travel cost.

	Coefficient	Clustered SE	z
Travel > 100 miles	-3.4841^{***}	0.0068	-513.2
Course speed (log course FE)	3.8501^{***}	0.0559	68.9
Pre-race qualifying probability	0.6920^{***}	0.0311	22.2
BQ pressure × course speed	-6.1005^{***}	0.1776	-34.4
BQ pressure × qualifying probability	-1.0603^{***}	0.0510	-20.8

Notes: McFadden conditional logit estimated by maximum likelihood. Each choice occasion is one non-Boston US marathon finish by a US-residence runner with at least one prior observed race; the choice set is the course actually run plus up to 9 other US courses drawn uniformly at random from the courses active in the same calendar year (sampled-alternatives estimator, consistent under uniform sampling). Standard errors are cluster-robust at the *runner* level (score-based sandwich with CR1 correction; 221,682 runner clusters, 299,938 occasions); runners contribute many occasions, so observations are not independent across occasions of the same runner. BQ pressure is occasion-constant, so its main effect differences out and it enters only through the interactions. Log-likelihood −492,046. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix H Course Speed Index

Table H1 reports the estimated course speed effect for all 1057 indexed marathons, sorted from fastest to slowest. Course effects are expressed as the log difference relative to Boston (the Empirical Bayes–shrunk AKM estimate), along with the percentage equivalent and minute conversions at two reference paces: 3:00 (the current M18–34 qualifying standard) and 3:20 (approximately the median Boston qualifier). Negative values indicate courses faster than Boston. The full table is available as a CSV file in the replication archive.

Table H1: Course Speed Index (Top 25 and Bottom 10). The column "Obs. Boston slow." reports the raw mean $\ln T_{\text{Boston}} - \ln T_{\text{origin}}$ in minutes at 3:20 pace from same-runner Boston-destination pairs (at least 50 pairs required; "—" otherwise). Under perfect additive separability, this number should equal the negative of the Min (3:20) column: a course 18 min faster than Boston should produce runners who are +18 min slower at Boston. Small gaps are consistent with measurement noise or Boston-specific factors; large gaps can indicate peaking (observed slowdown exceeds the FE prediction when the origin is a goal race) or course-measurement issues at the origin.

Marathon	Location	% vs Boston	Min (3:20)	Min (3:00)	Obs. Boston slow. (min)	Race-Years
25 Fastest Courses						
Inland Trail Marathon	Ohio	-10.2	-20.4	-18.3	+22.8	5
The Buckeye Marathon	Arizona	-9.9	-19.9	-17.9	+29.1	5
Hokkaido Marathon	Japan	-9.8	-19.5	-17.6	—	10
Abingdon Marathon	United Kingdom	-9.5	-19.0	-17.1	—	9
REVEL MT Charleston	Nevada	-8.5	-17.1	-15.4	+21.8	10
Adelaide Festival City Marathon	Australia	-8.5	-17.0	-15.3	—	23
Spring Chance Bq.2 Marathon	Illinois	-8.3	-16.6	-15.0	+10.3	6
Tunnel Lite Marathon	Nevada	-7.8	-15.6	-14.0	+19.1	11
Last Chance Bq.2 Chicagoland Marathon	Illinois	-7.7	-15.3	-13.8	+12.3	9
REVEL Big Bear and Half Marathon	California	-7.6	-15.1	-13.6	+21.7	8
Mountain To Surf Marathon	Czech Republic	-7.5	-15.1	-13.5	—	12
Last Chance Bq.2 Grand Rapids Marathon	Michigan	-7.2	-14.5	-13.0	+10.0	6
City Of Christchurch Marathon	New Zealand	-6.9	-13.9	-12.5	—	22
Light At The End Of The Tunnel Marathon	Washington	-6.4	-12.8	-11.5	+15.4	11
Canberra Marathon	Australia	-5.8	-11.6	-10.4	—	22
MT. Nebo Marathon	Utah	-5.7	-11.4	-10.3	+13.4	10
Carmel Marathon	Indiana	-5.6	-11.2	-10.1	+12.1	13
Hudson Mohawk Marathon	New York	-5.6	-11.2	-10.1	+14.9	10
Sunshine Coast Marathon	Australia	-5.5	-11.0	-9.9	—	9
Melbourne Marathon	Australia	-5.5	-11.0	-9.9	—	22
Townsville Marathon	Australia	-5.5	-11.0	-9.9	—	16
Newport Wales Marathon	United Kingdom	-5.5	-10.9	-9.8	—	5
Gold Coast Airport Marathon	Australia	-5.1	-10.3	-9.3	—	20
Brisbane Marathon	Australia	-5.0	-10.0	-9.0	—	10
Perth Marathon	Australia	-5.0	-10.1	-9.1	—	21
10 Slowest Courses						
Great Wall Marathon	United States	+37.2	+74.4	+66.9	—	9
Greenway Trail Marathon	Maryland	+38.3	+76.6	+68.9	-67.3	10
Bataan Memorial Death March	New Mexico	+39.4	+78.9	+71.0	-49.2	22
Aspen Backcountry Marathon	Colorado	+40.6	+81.2	+73.1	—	5
Six Foot Track (Ultra) Marathon	Australia	+42.8	+85.6	+77.1	—	6
Moab Trail Marathon	Utah	+42.9	+85.7	+77.2	-84.0	11
The North Face Endurance Challenge - Bear Mountain	New York	+46.5	+92.9	+83.6	—	9
Moose Mountain Marathon	Minnesota	+48.0	+95.9	+86.3	—	13
Leadville Trail Marathon	Colorado	+54.5	+109.1	+98.2	-103.8	22
Pikes Peak Marathon	Colorado	+76.0	+152.1	+136.9	-122.4	24